



Voluntary Carbon Standard
Project Description Template

19 November 2007

November 22nd, 2010

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List of Abbreviations

DSİ: General Directorate of State Hydraulic Works
EİE: General Directorate of Electrical Power Resources Survey and Development Administration
CDM: Clean Development Mechanism
JI: Joint Implementation
REC: Regional Environmental Center
UNFCCC: United Nations Framework Convention on Climate Change
EU: European Union
IEA: International Energy Agency
OECD: Organization for Economic Co-Operation and Development
CEE: Central and Eastern Europe
MENR: Ministry of Energy and Natural Resources
EÜAŞ: Electricity Generation Co. Inc.
EMRA: Energy Market Regulatory Agency
RER: Renewable Energy Resources
EML: Energy Market Law
ORKOY: General Directorate of Forest and Village Relations
TEİAŞ: Turkish Electricity Transmission Company
IPCC: Intergovernmental Panel on Climate Change
TİSK: Turkish Employer Unions Confederation

1 Description of Project:

1.1 Project title:

**Grid connected electricity generation from renewable sources:
Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey**
(Uzundere I HEPP, hereafter)

Document Version: 04

Date of Completion: November 22nd, 2010

1.2 Type/Category of the project:

The subject project is not a grouped project. According to domestic regulations, with an installed capacity of 63.0 MW the Uzundere I HEPP is qualified as a large scale project. The project comes under Type I – Renewable Energy Project as per Appendix B of the procedures for CDM project activities. The project is a 63.0 MW HEPP and it uses renewable sources to produce electricity. Since the installed capacity of the planned HEPP larger than 15 MW; it is a large scale renewable energy project activity according to the Decision 17/ CP.7 Article 6.

The expected annual average emission reduction is 87,318 tonnes as stated in Table 1 According to the VCS 2007:1 Guidelines section 5; the project falls into the individual validation and verification of a GHG programme option and the project size is “project” since the expected annual emission reduction is less than 1,000,000 tonnes and more than 5,000 tonnes.

1.3 Estimated amount of emission reductions over the crediting period including project size:

The project is not a grouped project. The crediting period of the project begins with the commissioning of the plant. The exact commencement date for the plant is the day of first documented supply to the grid with entire capacity. The expected annual electricity output and the estimated emission reduction along the first ten year of crediting are given below. Uzundere I HEPP is expected to generate 156,205 MWh electricity output until 2020. The starting date of the project activity is May 27th, 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine has been November 7th, 2010. The carbon crediting period and therefore the monitoring starts when the plant commences electricity generation. The crediting life is a maximum of ten years which may be renewed at most two times.

Year	Electricity Output MWh	Baseline Emissions (t CO ₂ e/yr)	Project Emissions (t CO ₂ e/yr)	Emission Reductions (t CO ₂ e/yr)
2010	58,202	32,534	0	32,534
2011	156,205	87,318	0	87,318
2012	156,205	87,318	0	87,318
2013	156,205	87,318	0	87,318
2014	156,205	87,318	0	87,318
2015	156,205	87,318	0	87,318
2016	156,205	87,318	0	87,318
2017	156,205	87,318	0	87,318
2018	156,205	87,318	0	87,318
2019	156,205	87,318	0	87,318
2020	98,003	54,784	0	54,784
TOTAL	1,562,050	873,180	0	873,180

Table 1: Estimated Amount of VCUs over the Crediting Period

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂e /yr)

PE_y = Project emissions in year y (t CO₂/yr)

LE_y = Leakage emissions in year y (t CO₂/yr)

The Project emissions from reservoirs of hydro power plants are calculated with the formula mentioned in ACM0002/Version11 as:

$$PE_{HP,y} = \frac{EF_{Res} \cdot TEG_y}{1000}$$

Where:

$PE_{HP,y}$ = Emission from reservoir expressed as tCO₂e/year

EF_{Res} = Is the default emission factor for emissions from reservoirs, and the default value as per EB23 is 90 Kg CO₂e /MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh).

The power density of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity, in W/m²

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of

the project activity, when the reservoir is full (m²)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero

$$Cap_{PJ} = 63,000,000$$

$$Cap_{BL} = 0 \text{ (Justification: The project is a new hydro power plant)}$$

$$A_{PJ}^1 = 7,275.54 \text{ m}^2$$

$$A_{BL} = 0 \text{ (Justification: The project is a new hydro power plant)}$$

Note regarding with A_{PJ} :

The project has two diversion weirs: Uzundere and Incesu. Since Uzundere is a tyrol type regulator, only the Incesu regulator area is included in the power density calculation.

For drinking water supply, irrigation or electricity generation purposes, water can be obtained from appropriate parts of rivers by an intake structure. Characteristics of river basins, hydrologic factors of regions and specifications and amount of sediment in rivers can affect type of water intake structure. The most commonly used water intake structure is “direct water intake”. In this structure, tyrol type regulators are used.

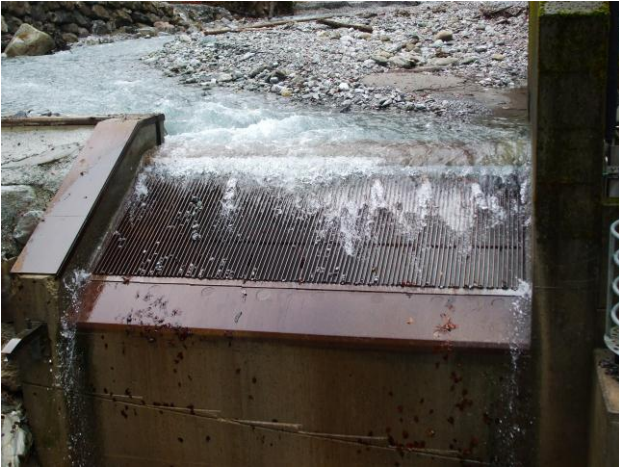
Picture 1 below demonstrates a tyrol type regulator structure. As it can be seen clearly in the picture, there is not any water holding upstream of the regulator. Water is obtained directly by a grate which is placed at water flow direction.



Picture 2: Tyrol type water intake structure

It also can be seen in Picture 2 in details. Water is not stored and obtained by the grate.

¹ *The project has two diversion weirs: Uzundere and Incesu. Since Uzundere is a tyrol type regulator, only the Incesu regulator area is included in the power density calculation.*



Picture 2 :Tyrol type water intake structure

Picture 3 and 4 are also examples for Tyrol type regulators.

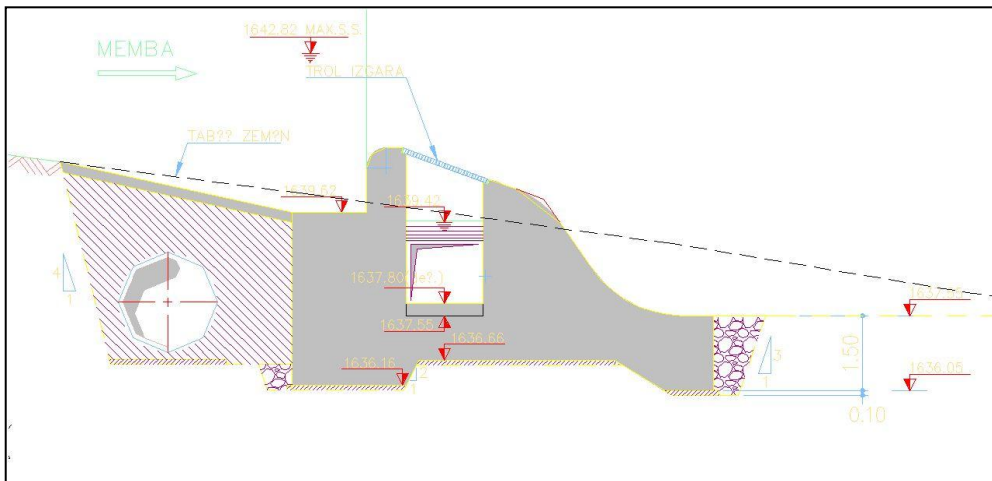


Picture 3



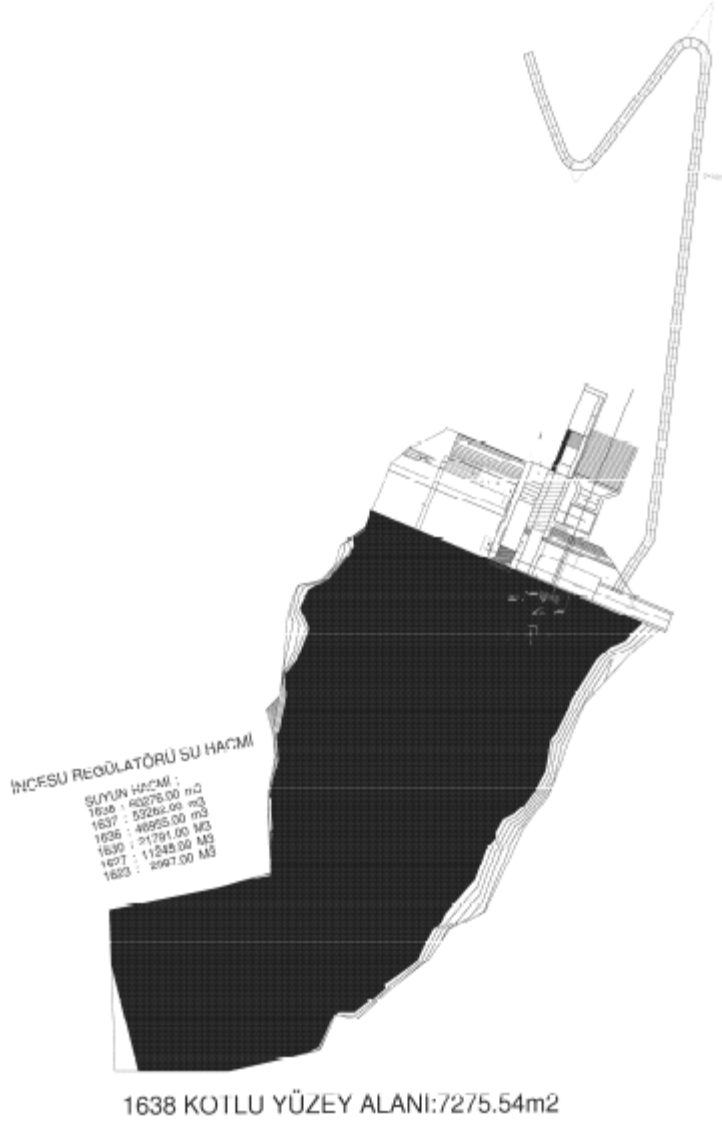
Picture 4

The water intake structure used in Uzundere part of Uzundere-I HEPP is tyrol type. The Uzundere part of the project is obtained water in the same way. There is no water holding at upstream of the regulator and water is obtained by the grate at the regulator area.



Picture 5 The intersection of the regulator at Uzundere Part of Uzundere-I HEPP

İNCESU REGÜLATÖRÜ MAX SU SEVİYE PLANI



Picture 6 The scheme of the regulator at Incesu Part of Uzundere-I HEPP used for the APJ which is 7,275.54

Therefore;

$$PD = (63,000,000 - 0) / (7,275.54 - 0) = 8,659.2 \text{ W/ m}^2$$

Therefore;

Since the Power Density of the Project is greater than 10 W/m² PE_y is assumed to be 0 as suggested in ACM0002 / Version 11.

The theoretical annual average energy production of the facility is based on the water flow data which was collected by “General Directorate of State Hydraulic Works” of Turkey which is the primary executive state agency of Turkey for Nations overall water resources planning, managing, execution and operation and “General Directorate of Electrical Power Resources Survey and Development

Administration (EİE)”. The water flow data which has been collecting by the DSİ AGİ (River Observation Station): Şenöz D. – Kaptanpaşa (DSİ 22-85) and EİE AGİ: İyidere – Şimşirli (Station No: 2218) was used to calculate the water amount the plant will be able to use. The annual average electricity production figure is based on the theoretical annual production figures with these historical annual water flows.

1.4 A brief description of the project:

Karadeniz Elektrik Üretim San. A.S. (KARADENİZ, hereafter) is in the phase of installing a large-scale runoff-river hydropower plant near Catal creek in Rize, Turkey. The project aims the installation of a 63 MW runoff-river hydropower electricity plant (HEPP).

The objective of the project is to generate electricity and supply it into the public grid Uzundere I HEPP will be registered as a Voluntary Emission Reduction (VER) project and facilitate the project implementation with carbon revenue coming from the carbon credits sale. Due to its significant contribution in diminishing carbon emissions and protecting the climate as well as due to some significant additionality issues discussed further below, this project is anticipated to fulfill VCS and qualify for carbon finance. The project has not been rejected by another GHG program before.

Before the start of the implementation of the project, the project area was largely forestry lands and KARADENİZ obtained all the permissions for these lands from the government. As for the rest of the project area, the company did not follow the standard appropriation process and purchased these private lands from the land owners with their consent.

In the baseline scenario, the electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants. Since Turkey’s grid mainly consists of thermal power plants, this would have resulted in GHG emissions. However, in the project scenario, the project whose characteristics are summarised below will generate electricity from hydropower and will result in emission reductions (relatively when compared to the baseline scenario) in parallel with its electricity generation figures.

There is no public funding available for the project.

1.5 Project location including geographic and physical information allowing the unique identification and delineation of the specific extent of the project:

Turkey is the host country. Turkey has ratified the Kyoto Protocol on February 5, 2009. Yet, in practice, Turkey will not commit to a cap on its greenhouse gas emissions and will not be a host for Clean Development Mechanism (CDM) or Joint Implementation (JI) projects until the end of 2012, because of its particular situation (Please see the final report of COP-7, Marrakesh)². The Ministry of Environment and Forestry is responsible with climate change policy making and climate protection project matters. The country office of the REC (Regional Environmental Center) in Turkey took the role of a National Focal Point for UNFCCC Article 6.

The project is located in the Black Sea Region of Turkey, within the province of Rize. The project is on the Uzundere creek. The altitude is 817.59 m at power house location. There is not any Nature Park, Natural Site Area or Protected Area at the project location. The project site had been vacant as the project is a new power unit and it did not replace an old plant.

The geographical locations of Uzundere I HEPP are 40° 50’ 16’’ - 40° 54’ 04’’ North and 40° 45’ 4’’ - 40° 49’ 42’’ East, respectively.

² http://ec.europa.eu/environment/climat/pdf/marrakech_report.pdf

UZUNDERE VE İNCESU WEIRS			
TUNNELS AND ART SRUCTURES			
(ED 50 DATUM - 3 DEGREE)			
ART STRUCTURE	X	Y	Z
İNCESU WEIR			
SAĞ KRET SINIRI	4523143,743	394667,232	
SOL KRET SINIRI	4523168,404	394607,084	
UZUNDERE WEIRS			
WEIR-1	4523931,486	401428,615	
	4523941,050	401412,193	
	4523939,459	401410,992	
	4523925,347	401410,554	
	4523947,451	401381,270	
	4523938,033	401366,367	
	4523935,288	401369,136	
	4523914,168	401407,106	
	4523914,985	401419,803	
WEIR-2			
	4523836,519	401276,156	
	4523826,586	401276,787	
	4523823,697	401266,173	
	4523832,396	401261,007	
	4523820,069	401261,455	
	4523818,752	401245,509	
	4523812,772	401246,004	
	4523814,090	401261,949	
	4523820,357	401280,555	
WEIR-3			
	4523852,368	401168,395	
	4523855,024	401139,110	
	4523858,661	401121,152	
	4523885,845	401123,772	
	4523885,653	401125,763	
	4523874,922	401132,766	
	4523862,011	401139,578	
	4523859,340	401169,027	
WEIR-4			

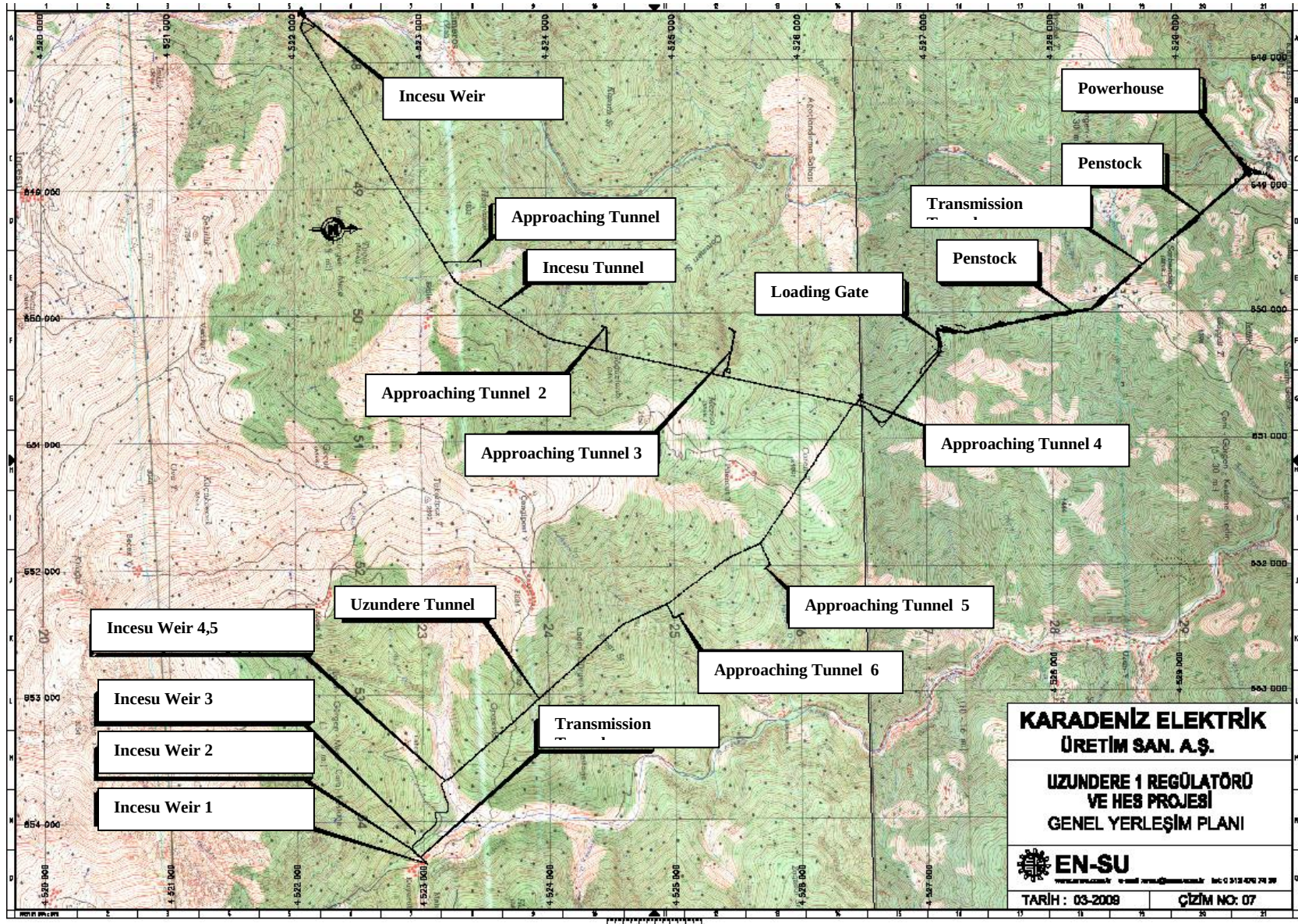


Figure 1: Site Layout



Figure 2: Project Location on Turkey Map

Implementation of the project will consist of construction of the following main items:

- Two diversion weirs, where water from the river is diverted into conveyance pipes;
- Total length of the conveyance tunnel is 11,324 m.
- Power house with Pelton type vertical turbines;

In the power house, two vertical shaft Pelton turbines will be installed, each with 31.6 MW (total capacity 63 MW). There are two generators attached to the facility. Generators have power factor of 0,9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

1.6 Duration of the project activity/crediting period:

The project reached a financial closure in November 2006.

Starting date of the project activity is May 27th, 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine has been November 7th, 2010. The carbon crediting period and therefore the monitoring starts when the plant commences electricity generation through the first turbine which started to supply electricity to the grid on May 27th, 2010. The crediting life is a maximum of ten years which may be renewed at most two times.

The first feasibility study for the project which was designed for 43 MW of installed power by Karadeniz was originally finalized in June 2005. After that, the detailed project design work was completed and search for financing of the project started in September 2006. In 2006, the license was granted for that first feasibility study. Contemporarily, it was figured out that it was possible to realize a new design for a 63 MW of installed power facility after detailed review of the water flow data and the new feasibility studies was started. Early 2007, the first civil works were started for the project. The same year, it was decided to increase the installed power to 63 MW by the board of KARADENİZ. Early 2008, the application for that increased installed power was made with a new feasibility study. Revised license was granted at late 2009.

The expected life time of the project is, starting from August 2006, minimum 49 years which corresponds to the duration of the licence obtained from EMRA. However, the equipments may be renewed (as recommended) in years according to the conditions of the equipments. The owner of the company will be able to renew the license, by applying to EMRA in 9 – 12 months before the expiration date of the first license duration.

1.7 Conditions prior to project initiation:

Prior to project initiation, the project area hosted some forestland and agricultural activities in small-scale lands (the most common crop in the region is tea). Because the land structure is broken and partite, agricultural economics has not developed in the region. Also, livestock production is not one of the main livelihoods in the region anymore, because it lost its value in time in parallel with the decrease in transhumance activities.

The project site had been vacant as the project is a new power unit and it did not replace an old plant.

1.8 A description of how the project will achieve GHG emission reductions and/or removal enhancements:

The existing electricity grid in Turkey is an interconnected single entity which to a larger extent fed by fossil fuel fired power plants. Uzundere I 63.0 MW HEPP Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources which otherwise would have been generated by the fossil fuel fired power plant.

The subject project will cause some GHG emissions during the operation as calculated in Section 1.3 of this document. The quantity of emissions created for the construction of the project is negligible according to ACM002 / Version 11.

1.9 Project technologies, products, services and the expected level of activity:

Uzundere I HEPP project entails the construction and operation of a 63.0 MW hydroelectric power plant. The plant consists of two turbines. The first-hand turbines which will be used in the project will be supplied from Alstom of which the lifetime is expected to be 35 years. The turbines will be Pelton type with vertical axis. The net electricity production (delivered to the grid after losses and consumption in the plant) from the plant is estimated to be 156,205 MWh per annum.

Input data of the project

Number and type of turbine	2 vertical pelton turbine
Water Flow per machine	4.75 m ³ /s
Downstream water level (maximum considered for design)	817.59 m.a.s.l.
Net head (1 unit in operation) [maximum head considered for design]	786.64 mWC
Net head (2 units in operation) [nominal head considered for design]	749.41 mWC
Waterhammer ace.	15%
Grid frequency	50 Hz
Runner material	Cr 13% -Ni 4% Stainless steel

Turbine

The vertical pelton turbine transforms the hydraulic energy into mechanical energy thanks to the pelton runner, transmitting the power to the generator shaft. The runner is by means of the turbine shaft, coupled to the generator shaft. In the upper part of the pelton runner, a guide bearing is placed in order to absorb all the radial efforts caused by the hydraulic effect on the pelton runner. The other two bearings of each unit are placed in the generator. The axial thrust caused by the weight of the mobile parts of the turbine and the hydraulic effect are absorbed by the upper generator thrust and guide bearing.

The pelton runner is perfectly water-fed thanks to the shape of the nozzle and to the inlet spiral distributor. The needle of the nozzle permits the gentle starting and stopping of the unit. For the emergency stop the deflector of each nozzle is used. Each needle is governed by an independent oil servomotor, placed inside of each needle, which is operated by the governing system of the unit. All deflectors are governed by a common oil servomotor and the effort transmitted to all the deflectors by means of levers. This servomotor is operated by the governing system of the unit. The housing is of the circular type and provided with removable screens allowing inspection and dismantling of the runner and nozzles. The housing air ventilation is obtained through pipes provided for this purpose. A decompression valve is provided in order to decompress collector and nozzles when the unit is standstill. This valve will also operate as air release valve.

Inlet Valve

The inlet valve is a spherical type main inlet valve and is provided for the isolation and the protection of the turbine from the upstream penstock. The valve is operated by means of one single acting oil operated servomotor, able to operate under the maximum head and against the maximum discharge. The closing of the valve is guaranteed by a counterweight. A security diaphragm is installed on the servomotor to secure

the closing of the inlet valve in the event of breaking of the oil pipe. One by-pass system permits the equalisation of the distributor and the penstock pressure before the opening of the inlet valve. It is operated by means of a double acting oil servomotor. Both servomotors are operated by means of solenoid pilot valves by using the turbine oil pressure unit. The dismantling of the inlet valve is done by means of a sliding loose flange and the watertightness is achieved by a joint which is compressed thanks to the sliding loose flange. The fund effect is transmitted to the penstock, so the penstock will have to be provided with a concrete anchorage out of the scope of ALSTOM Power Environment.

The valve will not be equipped with a double sealing system. When maintenance procedures are to be done in the unit, the penstock head valve, out of the scope of ALSTOM Power Environment, will be used in order to ensure the security in downstream part of the penstock and thereafter the isolation valve of each unit do not need to have a maintenance watertightness ensuring system. ALSTOM Power Environment could offer a double sealing system with the subsequent increase in the total cost.

Governing System, Speed Governor

Alstom proposed its digital speed Governor type TSLG, manufactured by ALSTOM POWER.

The TSLG assumes the regulation functions of the turbine.

Speed regulation.

The regulator will control the group in the start-up phase, driving it to the synchronous speed and allowing a faster coupling under the orders of a synchrocoupler (out of the ALSTOM Power Environment scope of supply). On the other hand, in the event of opening of the disjunct, the speed regulator will take the control of the group maintaining it in synchronous speed in waiting for the re-establishment of the net for a new coupling. Also, when the connection to the grid is not possible, it will operate in isolated grid, performing frequency regulation.

For a correct operation in isolated grid the necessary checks will be done (hydraulic and mechanical inertia of the groups, self-adjustment of the net and projected load steps).

Load regulation

The normal operation of the plant, will be connected in parallel to the grid. The turbine's participation degree in the frequency regulation of the grid will be function of the fixed statism in the TSLG.

The automaton of the plant will establish a value for the desired power. Depending on this set value, it will give the TSLG the opening or closing order. The command elements affected by the orders are the injectors that, in function of the demanded power of the group, operate or remain in closed position. The control loop is closed through the feedback of the position of the nozzles through the position continuous transducers (secondary regulation).

Governing System Specification

Scope of Supply

The oil governing system is made up:

- A hydraulic pressure unit (HPU),
- A hydraulic circuit to command the guard valve,
- A hydraulic circuit to command the nozzles and the deflectors,
- A cooling water system,

Note: the hydraulic station supports the pump, the command and control device of the hydraulic circuits and the securities.

Hydraulic Pressure Unit (HPU)

The hydraulic pressure unit is mainly composed of the following equipment:

- Two pumps, equipped with a check valve, supplies all the hydraulic circuits with pressurized oil.
- A pressure oil tank (with membrane) having sufficient capacity to perform the closing of the nozzles, in case of failing of the pumps. It permits also the intermittent working of the pump.
- A safety valve protected the circuit again over pressure.
- To guarantee a class of pollution compatible with the components used the HPU is insulated. An air vent equipped with a filter permits the circulation of air and the filtration of the oil during the tank filling. The pumps are provide with a suction filter and a filter pressure to protect the circuit.
- The control device of the HPU allows to manage the following parameters:
 - Oil level by the gauge, low level.
 - Oil temperature by the thermometer and the probe.
 - Oil pressure by the pressure gauge and pressure transducer.
- An electric terminal box regroups all the electrical input and output for all the components and device.

Guard Valve Circuit

The guard valve circuit is mainly composed of the following elements:

- The necessary pipes to convey the pressurised oil from the hydraulic station to the guard valve and by-pass.
- A electro valve to govern the servomotor which provides water-pressure to the sealing sliding ring of the guard valve.
- The by-pass and the control device to permit the equalise, up to a certain percentage, the inlet turbine pressure with the penstock pressure, before opening the guard valve. The solenoid pilot valve ensures the by-pass command. The diaphragm controls the operating times. Two limit switches, detect the bypass position completely opened-closed.
- A hydraulic cylinder open the valve. It closes using a counterweight. The distributor insures the guard valve command. The operating time is set using the controllers of debit. The diaphragm put on the hydraulic cylinder controls the closing time in case of pipe rupture.

The guard valve plug is equipped with switches that ensure the following functions:

- Detection of the closing, allows the pump stopping, during the stopping sequence, during the start-up sequence and the automatic start-up sequence.
- Opening detection, allows the by-pass closing.
- Detection of guard valve out of control, allows the stopping of the unit.

Nozzles and Deflector Circuits

The nozzles and deflector circuits is mainly composed of the following elements:

- The necessary pipes to convey the pressurized oil from the hydraulic station to the hydraulic cylinder

The nozzles are operated and closed using a hydraulic cylinder equipped of each nozzle by:

- One contact for the closing, one contact for open or closed the other nozzle and one contact to indicate the full opening nozzle.
- An inductive sensor to indicate the nozzle opening, providing a continuous indication.
- An electro-hydraulic valve type EDxxDE which is governed by a speed governor type TSLG (speed control system) insured the orders of operating and closing.

The operating times are insured by:

- Adjustable diaphragm for the closing
- Diaphragms put on the hydraulic cylinder for the opening,

The deflectors are operated using a hydraulic cylinder equipped with antagonist spring in order to ensure the closing of each deflectors:

- One contact to indicate the opening, another for the closing.
- An inductive sensor to indicate the deflectors opening.
- An electro-hydraulic valve type EDxxDE which is governed by a speed governor type TSLG (speed control system) insured the orders of opening and closing.
- Diaphragms put on the hydraulic cylinder for the closing.

The Project was designed by EN-SU Mühendislik Müşavirlik Ltd. Şti. under the supervision and control of DSI (Water State Works). The company has significant experience in the field of hydropower plant design, control and consultancy.

The technology used in the plant is a state-of-the-art technology with extensive automatization. In addition to this, the equipments which are used in the project are all firsthand equipments. There is no new technology which is expected to replace the plant's technology in the short run and the expected life of the equipment is around 20 years⁴. The project does not need extensive initial training and maintenance efforts in order to work as presumed during the project period. Necessary trainings to the plant staff will be provided by the supplier right after the installation of the electro mechanic equipments in line with the agreement between the project owner and the supplier.

The economic life of an HEPP in Turkey is assumed as 50 years³. However, it is recommended to renew the equipment in every 20 years⁴.

1.10 Compliance with relevant local laws and regulations related to the project:

It is important to note that there are no local laws and regulations in Turkey because it is not a federal state. The Uzundere I HEPP is in compliance with relevant national laws and regulations related to the project. The general scheme of the regulations this kind of facilities in line with, are defined by The Renewable Energy Law (Law Number: 5346, Acceptance Date: 10.05.2005) and Environmental Law (Law Number: 2872, Ratification Date: 09.08.1983).^{5,6} Such laws include land use regulations, forest use permissions and environmental risk management regulations.

1.11 Identification of risks that may substantially affect the project's GHG emission reductions or removal enhancements:

³ <http://ekutup.dpt.gov.tr/enerji/oik585.pdf> (page 4.25)

⁴ <http://ekutup.dpt.gov.tr/enerji/oik585.pdf> page 4.26

⁵ <http://www.epdk.gov.tr/mevzuat/diger/yenilenebilir/yenilenebilir.doc>

⁶ <http://www.cevreorman.gov.tr/yasa/k/2872.doc>

First of all, decrease of rainfall in the region due to the climate change, could reduce the GHG emission reduction capacity. However, recent studies show that region in which the plant will be commenced does not have a significant risk in comparison with the other regions in the country.⁷

In addition to that, the project is in the fourth degree seismic zone.⁸ Thus, the project does not have a considerable earthquake risk.

1.12 Demonstration to confirm that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction.

Not applicable. The project is not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction. The project is implemented as a Greenfield project and the purpose of the project is to generate electricity from hydropower without causing any GHG emissions.

1.13 Demonstration that the project has not created another form of environmental credit (for example renewable energy certificates).

There is no other form of environmental credit generated by the project because there is no such system within the country. The projects originate from Turkey do not comply for renewable energy certificates of EU because there is no energy trade between EU and Turkey because of different grid structures.

1.14 Project rejected under other GHG programs (if applicable):

Not applicable. The project was not rejected by any other GHG program.

1.15 Project proponents roles and responsibilities, including contact information of the project proponent, other project participants:

SUPPLIER DATA

Name	Karadeniz Elektrik Üretim San. A.Ş.
Organizational category	Private Company
Address	Fahrettin Kerim Gökay Cad. No:34 Altunizade 34662 Uskudar İSTANBUL-TURKEY
Main activities	Karadeniz Elektrik Üretim San. A.Ş. is a subsidiary of Eksim Group which is mainly involved with in renewable energy, real estate & construction, mining, food and commodity trading such as grain (wheat, soy, corn etc.) in Turkish domestic market and international market.
Contact	Mr. Oguz Tulgar +90 216 544 24 00 oguz.tulgar@eksim.com.tr Mr. Oguz Tulgar is the project coordinator of the Uzunder I Project, and responsible for the overseeing the carbon asset development activities on KARADENİZ side coordinating data flow to carbon asset development company (Gaia Carbon Finance), conducting relationships with the third parties such DOE, potential buyers etc.)

Table 2: Supplier Data

Name	GAIA Carbon Finance (GAIA)
Organizational category	Private Company

⁷ Bernhard Lehner, Gregor Czisch, Sara Vassolo, Energy Policy 33 (2005) 839–855, Page 851

⁸ <http://www.deprem.gov.tr/linkhart.htm>

Address	GAIA Finansal Danismanlik Hiz. Tic. Ltd. Sti Halaskargazi Cad. Zafer Sok. No: 11/4 Şişli 34371 ISTANBUL – TURKEY
Main activities	Carbon asset identification, development in Turkey. GAIA is the first domestic carbon asset development consulting company in Turkey.
Contact	Mr. Kagan Patir 0090 – 212- 224 04 50 kpatir@gaiacf.com Mr. Kagan Patir is the account manager of the Uzundere I Project on GAIA side, and responsible for preparing of the PDD, directing relationships with DOE on behalf of KARADENİZ, attending validation and verification activities on the side if necessary.

Table 2: Supplier Data

Karadeniz Elektrik Üretim San. A.Ş whose contact details given above is a project participant. Gaia Carbon Finance whose contact details given above is not a project participant.

1.16 Any information relevant for the eligibility of the project and quantification of emission reductions or removal enhancements, including legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and temporal information.):

The concept of hydropower projects undertaken by private sector is a very immature concept in the country. In this sense, the project will help Turkey to stimulate the commercial application of grid connected renewable energy technologies and markets. It will also be a crucial sample of establishing the feasibility of larger grid connected small to medium scale runoff-river HEPPs as a tool to support energy security, improved air quality, alternative sustainable energy, improved local source of income and sustainable renewable energy industry development. The specific benefits of the project are:

- Reduce greenhouse gas emissions in Turkey compared to the business-as-usual scenario,
- help to stimulate private sector participation in hydro power industry in Turkey,
- create employment during the construction and the operation phase of the plant,
- relatively reduce some other pollutants from power generation industry in Turkey, compared to a business-as-usual scenario,
- help to diminish Turkey's increasing energy deficit,
- diversify the electricity generation portfolio and reduce dependency on import of other energy sources.

Despite somewhat reasonable figures in the financial feasibility studies, bankability of renewable energy projects is still in question because banks and other financial institutions are very cautious in providing loans to these projects for various reasons: Lack of long-term markets in the country, poor understanding of private sector participation in renewable energy investment and the fragile macro-economic stability play a significant role in long-term financing decisions of financial institutions.⁹ Even if the projects are appraised to be bankable, the interest rates are significantly high due to risk premiums stemming from limited appetite of banks in providing loans to the industry. As private sector participation in renewable energy industry is a very new practice in the country, banks have limited experience and technical knowledge in the appraisal of such projects. The appetite for lending in such projects is curbed because the final output from the project in terms of electricity is always in question. Hence, confirmation of project existence from an independent third party and the resulting financial support from international

⁹ Hoffman, Scott L. The Law and Business of International Project Finance. Section 1-15. Cambridge Publications. August, 2007.

buyers of carbon emission would be a leap of faith for the banks and financial institutions which hesitate to be on the long-term financing side of such projects. The support would not only ease the crediting process for the project owners but would also give a stronger hand in negotiating for less expensive loans. Eventually, even if such projects achieve some financing from the local banks; the project owners are seldom content with the credit terms regarding the tenor and the price. As also seen in the subject Uzundere I HEPP project, there are also cases that the project developers face unexpected costs over the construction of the projects and seek extra financing to complete the project. Carbon finance revenue also becomes vital while rising extra financing for such projects.

The project contributes to sustainable development in Turkey in two major ways:

➤ Hydropower presents significant environmental benefits:

- Generating electricity from hydropower energy does not result in emissions of pollutants into the atmosphere with zero residuals that carry adverse impacts on soil, water etc. Regular emissions from conventional electricity generation such as sulphur dioxide, nitrogen oxide and particulates will not occur in this case.
- As a renewable energy source hydropower can be used without jeopardizing the supply of primary energy sources in the future
- The proposed project will significantly contribute to the reduction of GHGs.

➤ Hydropower presents significant economic benefits:

- The region is energy poor and the industry is undeveloped. But, thanks to its proximity to Black Sea markets, it carries enormous potential for industrial growth. Nevertheless, agriculture is destined to be the main livelihood in the region because of tea production and industry consists of only tea factories. Emission of pollutants such as sulphur dioxide through shifting to fossil fuel based electricity generation in the region to satisfy energy need of the region will result in damaging the quality of agriculture cause economic harm in the region.
- Construction of Uzundere I HEPP and operation of the plant will result in extra employment and had already provided many contributions in kind in the local area.
- During the construction of the project nearly 120 workers were employed periodically from the region and after the plant commenced, 10 people are being planned to work for the operation of the plant constantly.
 - The company tried to provide job opportunities to the local people as much as possible and 30% of the employees were hired from the region in the construction period. Also, the workers will be hired from the region in the operation of the plant in the ratio of 75%.
 - Despite the existing urgent appropriation decision with respect to the project which is a right granted by government via EMRA, all appropriations were made at market prices.
 - Construction materials for tunnels and all other equipment were supplied by the company through local resources all the way.
 - The electricity infrastructure of Catalder Village was renewed at the cost of 1 million Euro.
 - Village roads were maintained by the equipments and workers of the company. Thanks to this

contribution, it was made possible to transport from Cataldere Village to Incesu Village.

- The roads that were built for approach tunnels made possible forestry workers to intervene possible forest fires.
- The company provided financial support for the mosque in Cataldere Village.
- The Construction Site building will be left to the common use of Cataldere Village after the commissioning.
- The company provided its workers, equipments, and construction materials to different maintenance and repairmen works of Cataldere Village in many cases.
- In winter seasons, the company cleared up the closed roads of Cataldere Village.

1.17 List of commercially sensitive information

There is no any commercially sensitive information regarding with the project

2 VCS Methodology:

2.1 Title and reference of the VCS methodology applied to the project activity and explanation of methodology choices:

The baseline for the project was established through the official methodology of ACM0002 / Version 11, named “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”¹⁰ as approved by the CDM Executive Board. Also, to prove additionality and to calculate the grid emission factor, the official methodologies: “Tool for the demonstration and assessment of additionality, Version 05.2”¹¹ and “Tool to calculate the emission factor for an electricity system, Version 02”¹² are used. Conservative options and data were selected during the implementation of the methodology. The baseline has developed by Gaia Carbon Finance whose contact details are given in the Section 1.17. The completion date of the baseline study is 16/03/2010.

2.2 Justification of the choice of the methodology and why it is applicable to the project activity:

The choice of methodology *ACM0002 / Version 11* is justified as the proposed project activity meets relevant applicability criteria:

- Uzundere I 63 MW Runoff-River HEPP Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources and which supplies electricity to a system that is supplied by at least one fossil fuel fired generating unit,
- The geographic and system boundaries for the relevant electricity grid can be clearly identified and information on the characteristics of the grid is available as proved in section 2.4 “Description of how the baseline scenario is identified and description of the identified baseline scenario” of this PDD,
- The project activity is not implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m².

¹⁰ http://cdm.unfccc.int/EB/052/eb52_repan07.pdf

¹¹ <http://cdm.unfccc.int/EB/016/eb16repan1.pdf>

¹² <http://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-07-v2.pdf>

- Since the project activity involves a very small reservoir, it does not result in a power density of lower than 4 W/m², as per definitions given in the Project Emissions section.
- 5 years of historical data are available.
- The project does not involve switching from fossil fuels to renewable energy at the site of the project activity,
- The project involves construction of new units in a brand new plant, in other words the project does neither involve the addition of renewable energy generation units at an existing renewable power generation facility nor does it foresee to retrofit or modify an expired facility of renewable energy generation.

2.3 Identifying GHG sources, sinks and reservoirs for the baseline scenario and for the project:

Potential leakage emissions in the context of power sector projects are emissions that arise from the project activities such as power plant construction, fuel handling and land inundation. According to ACM0002 / Version 11, such emissions do not need to be taken into account.

Turkish Electricity Sector¹³

Demand Side:

Gross electricity demand (Gross Generation + Imports – Exports) in Turkey rose at a rate of 8.0% per annum for the last three decades since 1975, that reflects one of the basic indicators of an emerging economy. Gross electricity demand in Turkey reached 198,085 GWh in 2008, which makes Turkey one of the ten largest markets for electricity among European countries. Even the devastating earthquake that hit Turkey in 1999, and the economic crisis in 2001, could not reverse this robust growth record. In 1999, consumption grew by 4.5% despite the decline in GDP by 5%. In 2001, Turkey's electricity consumption dropped only 1.1% whereas GDP shrank by 7.5%. Parallel to the gross demand, the net consumption figure reflects an average of 7.9% increase since 1975, reaching to 161,948 GWh in 2008. Although there has been significant improvements in the recent years, Turkey's per capita electricity consumption figure is 2,053 kWh (as of 2006), which is one of the lowest among European countries.¹⁴ According to International Energy Agency (IEA) statistics, OECD average per capita electricity consumption is 8,381 kWh in 2006, which is an indicator of the growth potential in Turkey.¹⁵ This figure has been 3,258 kWh among CEE countries.¹⁶ Per capita consumption is expected to increase to 5,050 kWh by 2020 according to the MENR base case demand growth scenario, and with a projected annual population growth rate of 1.5%.

Supply Side:

Installed capacity and electricity generation in Turkey increased in line with consumption growth in recent decades. In 2008, electricity generation reached 198,418 GWh while imports stood at 789 GWh – 0.40% of total supply. Turkey became an importer in the mid-1970s, however, the share of imports in total supply never exceeded 3.6% for the last 20 years. The imports reached its peak as a portion of total supply in 1984 with 8%. By the end of 2008 the installed capacity of Turkey reached 41,817 MW, representing a CAGR of 8.0% since 1975. The rate of the increase in the capacity was well above 10% from 1970 to 1980. 48.2% of the installed capacity of 41,817 MW in 2008, was held by the State Generation Company EUAS, 9.2% by Affiliated Partnerships of EUAS, 0.6% Mobile Power Plants and the remaining 42.0% owned by autoproducers, Production companies, TOOR and ADÜAŞ. There is an increasing trend in favor of IPPs, switching away from the State Generation Company in the recent years. When we compare the breakdown of the installed capacity in 2000, we see that EUAS's contribution to the installed capacity was 80% whereas the IPPs' share was only 9%. This is a concrete indication of the trend of the generation in favor

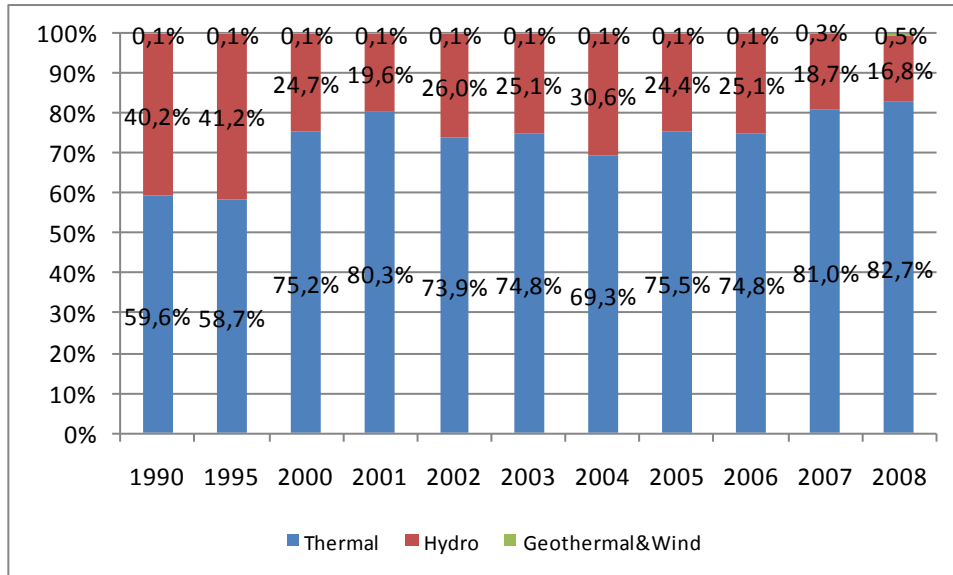
¹³ <http://www.teias.gov.tr/ist2007/index.htm>

¹⁴ http://www.iea.org/Textbase/country/maps/EUROPE/ele_pop.htm

¹⁵ http://www.iea.org/Textbase/stats/indicators.asp?COUNTRY_CODE=28

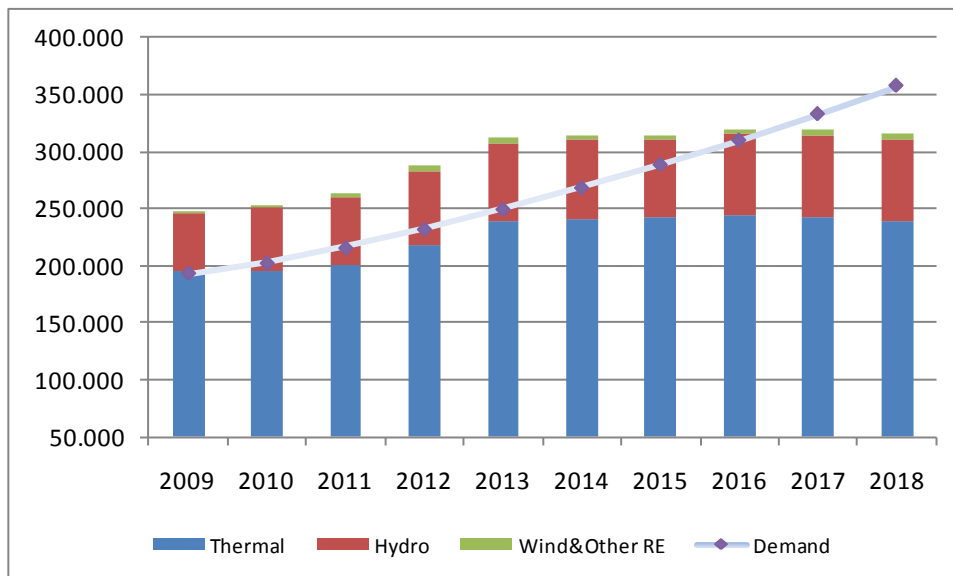
¹⁶ http://www.iea.org/Textbase/stats/indicators.asp?COUNTRY_CODE=33

of independent producers. The contribution of thermal power has always been the dominant source in both the generation and the installed capacity. In 2008, thermal power plants accounted for the 66.0% of the installed capacity whereas the hydro power plants' contribution was 33.1%. This is one of the lowest figures since 1970. On the other hand geothermal and wind powered capacity is still negligible with 0,9%. The existing electricity grid in Turkey is an interconnected single entity which to a larger extend fed by fossil fuel fired power plants. (Graph 2.1)



Graph 1: Electricity Production Fuel Type ¹⁷

The Graph below displays the forecast in increasing demand along the years between 2008 and 2017. The supply projection is based on the planned projects for energy production but it is very probable that, under increasing pressure of increasing demand, the government will shift to thermal power plants as they present higher financial feasibility with relatively short construction periods.



Graph 2: Electricity Supply and Demand Projections for Turkey¹⁸

Turkish Renewable Energy Sector

¹⁷ Source: [http://www.teias.gov.tr/istatistik2008/32\(75-08\).xls](http://www.teias.gov.tr/istatistik2008/32(75-08).xls)

¹⁸ Source: <http://www.teias.gov.tr/projeksiyon/KAPASITEPROJEKSIYONU2009.pdf>

Although Turkey is not a party to Kyoto Protocol (Minister of Forestry recently announced that Turkey will be a party to Kyoto Protocol under certain conditions), having signed the United Nations Framework Convention on Climate Change it is committed to managing greenhouse gas emissions. The goal is to maintain emissions at the level of 1990 (3,15 per capita). Therefore, Turkish government keeps focusing on renewables.

Turkey enacted its first specific renewable energy law in 2005, but there are also provisions regarding renewable energy in the Electricity Market Law (EML), which authorises the Energy Market Regulatory Authority ("**EMRA**") to take measures to promote renewable energy use and in secondary legislation:

- The Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electrical Energy¹⁹ ("**Law**") was adopted on 18 May 2005. This article provides as overview of the general provisions of the Law, as well as other supporting legislation regarding renewable energy resources. However, the legislation on bio-fuel will not be taken into account.²⁰
- Renewable energy resources ("**RER**") is not a brand-new topic, as the Electricity Market Law²¹ ("**EML**"), which was enacted in March 2001 and the Electricity Market License Regulation²² ("**Regulation**") demonstrate. According to the EML, EMRA is authorized to take the necessary measures to encourage the utilization of RER.
- According to the European Council Decision of 23 January 2006 on the principles, priorities and conditions contained in the Accession Partnership with Turkey²³, one of the short-term priorities identified for Turkey relates to "[s]tart alignment on the *acquis* on energy efficiency and renewable energy sources and develop administrative capacity in these sectors". Therefore, the new Law contains new incentives for RER development.

Objective and Scope of the Law

It is important to note that there is no region specific regulation and law in Turkey as it is not a federal state. The purpose of the Renewable Law is "to expand the use of renewable energy resources for generating electricity and to benefit from these resources in a secure, economic and qualified manner; and to increase the diversification of energy resources, reduce greenhouse gas emissions, assess waste products, protect the environment and develop the related manufacturing sector to realize these objectives."

The Renewable Law covers wind, solar, geothermal, biomass, biogas, wave, stream, tidal, river and arc type hydroelectric generation facilities and hydroelectric generation facilities either canal or run of river type or with a reservoir area of less than 15 km².

Incentives provided by the Renewable Law and Electricity Market Licensing Regulation

Renewable Energy Law

(a) Development plans which might have a negative effect on the use and efficiency of RER areas can no longer be created on public land.

(b) Each legal entity holding a retail sale licence must purchase a specified amount of electrical energy from RER certified generators which have been in generation for less than 10 years. This amount is based on a comparison between the amount of energy sold by that retail sale licence

¹⁹ Law No. 5346, published in the Official Gazette dated 18 May 2005 and numbered 25819.

²⁰ See Petroleum Market Law No. 5015, published in the Official Gazette dated 20 December 2003 and numbered 25322; Petroleum Market License Regulation, published in the Official Gazette on 17 June 2004 and numbered 25495; Regulation on Technical Criteria for Petroleum Market, published in the Official Gazette on 10 September 2004 and numbered 25579.

²¹ Law No. 4628, published in the Official Gazette dated 3 March 2001 and numbered 24335

²² Published in the Official Gazette dated 4 August 2002 and numbered 24836; see moreover Electricity Market Grid Regulation, published in the Official Gazette dated 22 June 2003 and numbered 25001; Regulation on Balancing and Settlement, published in the Official Gazette dated 21 December 2004 and numbered 25677; Communiqué Regarding the Principles and Procedures of Financial Settlement, published in the Official Gazette dated 4 November 2003 and numbered 25279.

²³ OJ 2006 L 22/34.

holder, in the previous calendar year, and the total electrical energy offered for sale by all retail sale licence holders in Turkey.

The price of electrical energy bought in accordance with this provision is determined by EMRA and is the average Turkish wholesale price announced in the previous year. This amount is 9.67 Ykr/Kwh in 2007 (approximately 5 Euro cents). The retail price must be between 5.0 and 5.5 Euro cents but a generator can sell its electrical energy for a higher price if there is market demand.

In practice all generators are currently selling their electrical energy to the Market Financial Reconciliation Center, which currently offers the highest price in Turkey due to a recent supply gap.

(c) Real persons and legal entities establishing an isolated electricity generation plant and grid supported electricity generation plant; using hydraulic resources with a maximum installed capacity of 1,000 kW that is to be used solely to satisfy their own needs, are not required to pay service charges for these projects. This is provided that the final design, planning, master planning, preliminary surveying and first auditing were prepared by either the DSI (State Hydraulic Works) or the EIE (Electrical Power Resources Survey and Development Administration).

(d) The sale price, rent, rights of access and usage permissions of state owned land are subject to an 85% reduction where the property is used for the purpose of generating electrical energy from RER which fall within the scope of the Renewable Law. ORKOY (General Directorate of Forest and Village Relations) and forestation special allowance revenue are not charged for forested land.

(e) Within the framework of the Renewable Law: (i) investment in energy generation facilities; (ii) procurement of domestically manufactured electromechanical systems; (iii) investment in research, development and manufacturing in the scope of electricity generation systems using solar cells and concentrated collectors; and (iv) investment in research and development facilities for the generation of electrical energy or fuels by utilizing biomass resources, can benefit from incentives determined by the Council of Ministers. Nevertheless, despite good intention of the government to promote electricity generation from renewable energy sources, it is not possible to say that the incentives provided so far are sufficient or the existing incentives are applied properly.

Licensing Regulation

Legal entities applying for Licences for the construction of facilities based on domestic natural resources and RER only pay 1% of the total licensing fee and do not pay annual Licence fees for the first 8 years following completion of the facility.

Legal entities engaged in generation activities at facilities based on RER can purchase electricity from private sector wholesale companies on the condition that they do not exceed the annual average generation amounts indicated in their Licences for that calendar year.

TEIAS and/or legal entities holding distribution Licences must give priority for connection of generation facilities to the system based on whether they use domestic natural resources and RER.

Licence guarantee

Since November 2007 EMRA has been requesting bid bonds and performance bonds for Licence applications. Bid bonds with an amount of 10,000 YTL per mw are requested at the application stage and, if EMRA approves the application, a performance bond, with an amount to be calculated in accordance with the capacity to be installed, is also requested.

Renewable Energy Resource (RER) Certificates

If and when requested by any legal entity holding a generation Licence, a RER certificate must be granted by EMRA for the purpose of identifying and monitoring the resource type used in electrical energy generation or for the purpose of accessing the incentives applicable under the Renewable Law.

The principles and procedures relating to RER certification are governed by secondary legislation issued by EMRA in 2005. There are two types of RER certificates which govern the:

- (a) type of resource utilized to generate electrical energy; and
- (b) incentives that the owner is entitled to under the Renewable Law.

No GHG sources in terms of sinks and reservoirs could be identified for this project.

2.4 Description of how the baseline scenario is identified and description of the identified baseline scenario:

The Uzundere I Hydroelectric Power Plant Project entails the construction and operation of a 63 MW hydroelectric power plant. The project consists of 2 diversion weirs with turbines of two 31.6 MW.

	Source	Gas	Included?	Justification
Baseline	Generation mix of electricity grid in Turkey	CO ₂	Yes	CO ₂ emission from fossil fuel fired power plant that are displaced due to project activity was taken into account.
		CH ₄	No	Minor emission sources as suggested by ACM0002, Version 11
		N ₂ O	No	
Project Activity	Construction and Operation of the project activity	CO ₂	No	As net electricity approach is adopted, emissions that occur during construction and during the operation are negligible and non-existent respectively.
		N ₂ O	No	Minor emission source as suggested by ACM0002, Version 11
		CH ₄	No	Minor emission source as suggested by ACM0002, Version 11. Emissions that occur during the construction are negligible.

Table 3: The justification for project boundary

The method to describe and calculate the baseline has clearly been specified by the Baseline Methodology. CDM Executive Board has already provided a consolidated tool for appraising and demonstrating the additionality feature of the projects.

Since the project is an installation of a new grid-connected renewable power plant, the baseline scenario is formulated in ACM0002 / Version 11: *“Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the Combined Margin (CM) calculations described below”*.

The project is not a modification/retrofit of an existing grid-connected renewable power plant/unit. Therefore the other alternative baseline scenario mentioned in ACM0002 / Version 11 is not chosen.

According to the “Baseline Methodology Procedure” in “Tool to calculate the emission factor for an electricity system, Version 02” following steps should be followed.

All the information pertaining to the grid and estimating baseline emissions are publicly available, and was available at the website of TEIAS (Turkish Electricity Transmission Company Inc. - www.teias.gov.tr).

STEP 1: Identify the relevant electricity system

The project electricity system is defined as “the spatial extent of the power plants that are physically connected through transmission and distribution lines to the project activity (e.g. the renewable power plant location or the consumers where electricity is being saved) and that can be dispatched without significant transmission constraints” in the “Tool to calculate the emission factor for an electricity system, Version 02” page 3. Similarly, “a connected electricity system, e.g. national or international, is defined as an electricity system that is connected by transmission lines to the project electricity system. Power plants within the connected electricity system can be dispatched without significant transmission constraints but transmission to the project electricity system has significant transmission constraint” in the same document.”

In the project’s case “the project electricity system” and “the connected system” are same. As also confirmed by TEIAS (Turkish Electricity Transmission Company Inc.), the Turkish transmission system is interconnected. There is an independent regional grid system neither in Rize nor in Black Sea Region.

In addition to this, since there is no DNA in the host country to delineate the project electricity system, the suggested criteria in “Tool to calculate the emission factor for an electricity system, Version 02” was used. According to this, 1-The capacity usage figure for the transmission line should be checked. 2-Spot market prices of different systems in the country should be compared.

Since there is no capacity usage figure for transmission line published, the criteria “The transmission line is operated at 90% or more of its rated capacity during 90% percent or more of the hours of the year.” could not be proved.

On the other hand, there is no spot electricity market available in the country as suggested in the other criteria “In case of electricity systems with spot markets for electricity: there are differences in electricity prices (without transmission and distribution costs) of more than 5 percent between the systems during 60 percent or more of the hours of the year.” Therefore, this criterion is not applicable as well.

As suggested in “Tool to calculate the emission factor for an electricity system” Version 02, “if these criteria do not result in a clear grid boundary, use a regional grid definition in the case of large countries with layered dispatch systems (e.g. provincial / regional / national).” However, there are no layered dispatch systems in the host country. As a result the national grid was used as the project electricity system. Hence, the estimation of OM (Operating Margin) and BM (Built Margin) are based on the definition of the Turkish electricity network as one single interconnected system.

STEP 2: Choose whether to include Off-grid power plants in the project electricity system

According to “Tool to calculate the emission factor for an electricity system, Version 02”, the OM and BM values can be calculated based on the following two options:

- Option I: Only grid power plants are included in the calculation
- Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I is selected in the calculation of Built Margin and Operating Margin for the proposed project activity. The reason why the tool suggests the Option II is, “in some countries off-grid power generation is significant and can partially be dispatched by CDM project activities, e.g. if off-grid power plants are operated due to an unreliable and unstable electricity grid.”

As Turkey’s grid is more appropriate for calculating the OM and BM values according to Option I, this option is selected.

STEP 3: Select a method to determine the operating margin (OM)

According to “Tool to calculate the emission factor for an electricity system, Version 02”, four alternative methods to calculate OM can be chosen. In choosing the right method for the calculation of OM, “Simple

adjusted OM”, “Dispatch data analysis OM” and “Average OM” methods are eliminated since all these methods require power plant specific information of power plants which are connected to the grid. However, no power plant specific information is available. Similarly, option A of “Simple OM” method was also eliminated as again since there is no publicly available power plant specific data in Turkey.

Simple OM Option B can only be used if:

- (a) The necessary data for Option A is not available; and
- (b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
- (c) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

The only relevant low operating cost and must run resource in Turkey is hydropower because the share of all other renewable resources is close to nil and therefore can be assumed negligible. The share of all non-hydro renewable resources in the entire electricity generation is 0,5 percent for the years between 2000 and 2008. There is no example of coal being used as must-run and nuclear energy is not practiced in Turkey. As depicted in Table 4²⁴ the share of low-cost/must run resources were never higher than 50 percent between 2000 and 2008. Also, according to the TEIAS statistics, the share of hydroelectricity production of Turkey has not exceeded 50% for 20 years (since 1989).²⁵

	2000	2001	2002	2003	2004	2005	2006	2007	2008
Turkey's Gross Electricity Production (GWh)	124.922	122.725	129.400	140.581	150.698	161.956	176.300	191.558	198.418
Electricity Production From Hydro (GWh)	30.879	24.010	33.684	35.330	46.084	39.561	44.244	35.851	33.270
Share (%)	25%	20%	26%	25%	31%	24%	25%	19%	17%

Table 4: Share of hydroelectric production in Turkey, 2000 – 2008

All in all, option B of “Simple OM” method was adopted as “the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system” is the only available data in the host country.

Step 4: Calculate the operating margin emission factor according to the selected method

The Simple Operating Margin Emission Factor ($EF_{OM,y}$) is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must run power plants/units as determined by the Baseline Methodology.

According to “Tool to calculate the emission factor for an electricity system, Version 02”, the formula given below is applied for computing the $EF_{grid, OMsimple, y}$.

Option B – Calculation based on total fuel consumption and electricity generation of the system:

$$EF_{grid, OMsimple, y} = \frac{\sum_i (FC_{i,y} \times NCV_{i,y} \times EF_{CO_2, i, y})}{EG_y}$$

Where:

$$EF_{grid, OMsimple, y} = \text{Simple operating margin CO}_2 \text{ emission factor in year } y \text{ (tCO}_2\text{/MWh)}$$

²⁴ All data presented in the baseline calculations are provided from official sources as mentioned in related footnotes. Therefore, uncertainties of data sets were not estimated.

²⁵ [http://www.teias.gov.tr/istatistik2008/32\(75-08\).xls](http://www.teias.gov.tr/istatistik2008/32(75-08).xls)

$FC_{i,y}$	=	Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fossil fuel type i in year y (GJ / mass or volume unit)
$EFCO_{2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ /GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost / must-run power plants / units, in year y (MWh)
i	=	All fossil fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

The fuel consumption values ($FC_{i,y}$) for each fossil fuel type are taken from the TEIAS statistics. Also, the net calorific values ($NCV_{i,y}$) are derived from these statistics by dividing total calorific values of fossil fuel types (GJ) by total amount of fossil fuel consumed in that type of power plants (tonnes).

In order to calculate the emission amounts from each fuel types, emission factors from IPCC website were used.²⁶ Three sets of emission factors were identified as minimum, medium and maximum values. To be conservative side, the minimum values are used in the OM calculations.

	kg CO ₂ /GJ			Default Carbon Oxidation Factor
	min	mid	max	
hard coal	92,8	96,1	100,0	1,0
lignite	90,9	101,0	115,0	1,0
fuel oil	75,5	77,4	78,8	1,0
diesel oil	72,6	74,1	74,8	1,0
natural gas	54,3	56,1	58,3	1,0
lpg	61,6	63,1	65,6	1,0
naphta	69,3	73,3	76,3	1,0

Table 5: Emission factors from IPCC

Turkey's GHG Emissions Inventories for year 2006 and 2007 are announced by Turkish Statistical Institute (TUIK). However, to be on the conservative side, CO₂ emissions figures from electricity production are not taken from these announced figures, as the calculation method is not known and these figures are a bit higher than the calculated figures.^{27, 28, 29} As a result, for years 2006, 2007 and 2008, the CO₂ emissions are calculated with the IPCC minimum values:

	2006	2007	2008
CO₂ Emission from Electricity Production (tons)	83.173.585	98.352.660	104.062.368

Table 6: CO₂ Emission of Turkey from Electricity Production

The calculation of net electricity production is demonstrated below.³⁰ As the efficiency factor from gross to net electricity for thermal resources is not known, the overall relation between gross and net electricity production is assumed to be the same for thermal production. Table 7 shows the overall gross/net relation

²⁶ http://www.ipcc-nggip.iges.or.jp/EFDB/find_ef_s1.php

²⁷ <http://www.tuik.gov.tr/PreHaberBultenleri.do?id=1996>

²⁸ <http://www.tuik.gov.tr/PreHaberBultenleri.do?id=4078>

²⁹ http://www.tuik.gov.tr/PreIstatistikTablo.do?istab_id=488

³⁰ [http://www.teias.gov.tr/istatistik2008/30\(84-08\).xls](http://www.teias.gov.tr/istatistik2008/30(84-08).xls)

where the estimated net electricity production from thermal resources were calculated by using the same relation.

	2006	2007	2008
Gross Electricity Production (a)	176.300	191.558	198.418
Net Electricity Production (b)	169.543	183.340	189.762
Net/Gross (c= a/b)	0,962	0,957	0,956
Gross Electricity Production from thermal sources (d)	131.835	155.196	164.139
Net Electricity Production from thermal sources (c*d)	126.783	148.538	156.979

Table 7: Net Electricity Production of Turkey from Thermal Sources

Obviously, using the same relation for both overall electricity production and thermal production is an approximation based on a rough assumption. Yet, obviously, such assumption results in a very conservative estimation because the efficiency of thermal plants is much lower than other plants as the operational consumption in those plants are relatively higher and this would only lead to a lower net electricity generation with higher OM emission factor and higher emission reductions.

With respect to the Baseline Methodology, electricity import amount is added to the domestic supply where the imports from connected grids located in other countries are weighted with an emission factor of zero tCO₂/MWh.

The last part of Step 1 is calculating the ratio of emissions and generation as follows:

	2006	2007	2008
OM (MWh/tCO₂)	0,657	0,663	0,664

Table 8: OM Emission Factor for 2006 – 2008

According to the methodology, two data vintages are allowed for the calculation of the OM emission factor, ex-ante and ex-post:

- Ex-ante: A 3-year average, based on the most recent statistics available at the time of PDD submission
- Ex-post: The year in which project generation occurs, if the OM emission factor is updated based on ex-post monitoring

In this case, the ex-ante approach is preferred because the data is available and it is a conservative approach due to the forecast that the weight of fossil fuel use in the generation of electricity in Turkey will increase.

As the weighted average of the figures between 2006 and 2008 is computed, the OM emission factor is 0,661 tCO₂ / MWh.

STEP 5: Identify the group of power units to be included in the build margin

According to the Baseline Methodology, the Build Margin (BM) Emission Factor EF_{BM} is calculated as the generation-weighted average emission factor of a sample of power plants m for a specific year, as follows:

$$EF_{grid,BM,y} = \frac{\sum_{i,m} EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

$EF_{grid,BM,y}$	Build margin CO ₂ emission factor in year y (tCO_2 / MWh);
$EG_{m,y}$	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	CO ₂ emission factor of power unit m in year y (tCO_2/MWh)
m	Power units included in the build margin
y	Most recent historical year for which power generation data is available

There is no available CO₂ emission data for the power units included in the BM calculation. The available data for on plant basis is the name, type, installed capacity and annual electricity production figures of these facilities. However, there is no available data for the electricity production figures of some small facilities, most probably because these projects are included in the “Other Projects”. Therefore, annual electricity productions of these facilities were calculated with the following formula unless the exact production figures are reached. The data used in the BM calculation is provided from the Turkey’s Capacity Projection Reports published in years: 2009³¹, 2008³², 2007³³, 2006³⁴ and 2005³⁵.

$EG_{m,y}$ = Full Load Working Hours x Installed Capacity

In addition to this, the efficiency rates for the fuel consumption of these facilities should be added to the calculation (as the thermal plants do not consume 100% of the fuel which they are fed with). Within the context of the calculation in this document, higher the efficiency figure means lower the CO₂ emissions (as the electricity calculation assumes that the electricity production is a function of the designed installed capacity not the efficiency. However, in theory the installed capacity of a plant increases with the higher efficiency).

Therefore, in the calculation of the $EF_{grid,BM,y}$, first $EF_{EL,m,y}$ values are calculated by using the formula (4) as suggested in the tool. Also, to calculate the $EF_{EL,m,y}$ values, the default efficiency figures listed under the “Appendix-1: Default efficiency factors for power plants” are used.

31 <http://www.teias.gov.tr/projeksiyon/KAPASITEPROJEKSIYONU2009.pdf>

32 <http://www.teias.gov.tr/projeksiyon/KAPASITEPROJEKSIYONU2008.pdf>

33 <http://www.teias.gov.tr/projeksiyon/KAPASITE%20PROJEKSIYONU%202007.pdf>

34 <http://www.teias.gov.tr/projeksiyon/KAPASITE%20PROJEKSIYONU%202006.pdf>

35 <http://www.teias.gov.tr/projeksiyon/KAPASITE%20PROJEKSIYONU%202005.pdf>

Grid Power Plants		
Generation Technology	Old Units (before and in 2000)	New units (after 2000)
Coal	-	-
Subcritical	37%	39%
Supercritical	-	45%
Ultra-supercritical	-	50%
IGCC	-	50%
FBS	35,5%	-
CFBS	36,5%	40,0%
PFBS	-	41,5%
Oil	-	-
Steam turbine	37,5%	39%
Open cycle	30%	39,5%
Combined cycle	46%	46%
Natural Gas	-	-
Steam turbine	37,5%	37,5%
Open cycle	30%	39,5%
Combined cycle	46%	60%

Table 9: Default Efficiency Factors for power plants (Appendix 1 of the tool)

As seen in the table above, the efficiency figures of coal, oil and natural gas, for different generation technologies, are given in the tool. In Turkey, the generation technologies for natural gas and oil is “Combined cycle” and for coal “Subcritical”. Since the default values for lpg and naphta are not given in the tool, to be on the conservative side, the efficiency factors of these fuel types are assumed as 60% (equal to the highest efficiency figure / Efficiency figure for Natural Gas-Combined Cycle).

As a result, in line with the tool, the efficiency factors for each fuel type are assumed as presented in the table below:

Fuel Type	Generation Technology	Efficiency Factor
hard coal	Coal - Subcritical	39%
lignite	Coal - Subcritical	39%
fuel oil	Oil - Combined Cycle	46%
diesel oil	Oil - Combined Cycle	46%
natural gas	Natural Gas - Combined Cycle	60%
lpg	Natural Gas - Combined Cycle	60%
naphta	Natural Gas - Combined Cycle	60%

Table 10: Efficiency Factors used for BM Emission Factor Calculation

The Build Margin estimation is based on the emission factors calculated for each energy source.

Fuel Specific Emission Factors

Carbon content factors for each fuel type (tC/TJ) refer to the factors stated in “2006 IPCC Guidelines for National Greenhouse Gas Inventories” and the fuel specific CO₂ emission factors were estimated from the same factors. To be on the conservative side, minimum emission factors were used in the calculation.

Electric Efficiency Rates

Default efficiency factors given in the tool. (Please see table 10)

Equivalent Electricity Generation

The equivalent electricity generation for each fuel type j out of the most recent 20 percent plants is calculated as follow:

Average Running Hours (Hours): For each specific fuel type, average running hours of similar facilities in Turkey for the year 2008, 2007, 2006 and 2005. Relevant figures were calculated based on the installed capacity and electricity production figures. On the other hand EUAS announces the electricity production figures of its facilities in each year. For these facilities actual figures were used.

Installed Capacity (MW): Total installed capacities of facilities run by similar fuel types for the period between January 8th, 2004 and December 31st, 2008.

Electricity Production (GWh): The announced electricity production figures of the facilities which were commissioned in the period between January 8th, 2004 and December 31st, 2008. And, multiplication of running hours and installed capacity for each fuel type, unless the energy production figures are not reached.

The computations display a weighted Build Margin emission factor of 0.457 tCO₂ / MWh.

STEP 7: Calculate the combined margin emissions factor

The baseline emission factor is the weighted average of the Operating Margin Emission Factor and Build Margin Emission Factor. The ACM0002 / Version 11 guideline recommends equal weight values for hydropower projects as seen in the formula below:

$$EF_{grid, CM, y} = EF_{grid, OM, y} \times w_{OM} + EF_{grid, BM, y} \times w_{BM}$$

Where:

$EF_{grid, BM, y}$	=	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EF_{grid, OM, y}$	=	Operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
w_{OM}	=	Weighting of operating margin emissions factor (%)
w_{BM}	=	Weighting of build margin emissions factor (%)

The default values recommended by Methodological tool: “Tool to calculate the emission factor for an electricity system, Version 02” for w_{OM} and w_{BM} for activities other than wind and solar power generation projects are 0.5 and 0.5, respectively.

$$EF_{grid, CM, y} = 0.661 \times 0.5 + 0.457 \times 0.5$$

Therefore resulting $EF_{grid, CM, y}$ is 0.559 tCO₂/MWh³⁶

The emission reductions of the project are calculated based on the formula shown below

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂e /yr)

PE_y = Project emissions in year y (t CO₂/yr)

³⁶ This value is given by being rounded down. Please see Annex 3 to see the exact values of OM, BM and Combined margin Emission Factors in 7 decimals.

LE_y = Leakage emissions in year y (t CO₂/yr)

Project emissions

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y =	Project emissions in year y (tCO ₂ e/yr)
$PE_{FF,y}$ =	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)
$PE_{GP,y}$ =	Project emissions from the operation of geothermal power plants due to the release of Non-condensable gases in year y (tCO ₂ e/yr)
$PE_{HP,y}$ =	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)

$PE_{FF,y}$ and $PE_{GP,y}$ are both irrelevant with the project activity and therefore assumed “0”, as the proposed project activity is a new grid-connected run-off-river hydro power plant.

On the other hand, ACM0002 Version 11 suggests that project proponents shall account for CH₄ and CO₂ emissions for the reservoir. Although the project does not have a reservoir and result in only a small lake which is attached to the regulator of the facility, the proposed calculations were run to prove the fact that the project’s emissions can be assumed “0”.

The Project emissions are calculated with the formula mentioned in ACM0002 / Version11 as:

$$PE_{HP,y} = \frac{EF_{Res} \cdot TEG_y}{1000}$$

Where:

$PE_{HP,y}$ =	Emission from reservoir expressed as tCO ₂ e/year
EF_{Res} =	Default emission factor for emissions from reservoirs of hydro power plants in year y (CO ₂ e /MWh)
TEG_y =	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh).

However, again according to ACM0002 / Version 11 if the power density (PD) of the hydro power plant is above 10 W / m², PE_y is 0.

The power density of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD =	Power density of the project activity, in W/m ²
Cap_{PJ} =	Installed capacity of the hydro power plant after the implementation of the project Activity (W)
Cap_{BL} =	Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero
A_{PJ} =	Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m ²)
A_{BL} =	Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²). For new reservoirs, this value is zero

$Cap_{PJ} = 63,000,000$
 $Cap_{BL} = 0$ (Justification: The project is a new hydro power plant)
 $A_{PJ} = 7,275.54 \text{ m}^2$
 $A_{BL} = 0$ (Justification: The project is a new hydro power plant))

Therefore;

$$PD = (63,000,000 - 0) / (7,275.54 - 0) = 8,659.2 \text{ W/ m}^2$$

Therefore;

Since the Power Density of the Project is greater than 10 W/m^2 PE_y and LE_y are assumed to be 0 as suggested in ACM 0002 / Version 11.

Accordingly the baseline emissions BE_y are calculated as following:

$$BE_y = EG_y \times EF_{\text{grid, CM, y}}$$

Where:

BE_y Baseline emissions in year y (tCO₂e/yr)
 EG_y Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
 $EF_{\text{grid, CM, y}}$ Baseline emission factor (tCO₂e/MWh)
 Y Refers to a given year

As a result, the expected baseline emission for the full year production of the project is:

$$BE_y = 156,205 \times 0.559 = 87,318 \text{ tCO}_2\text{e}$$

In the baseline scenario, Uzundere-I HEPP would not be constructed. Therefore, in the absence of the proposed VCS project activity, no other technology would be employed and the electricity amount to be generated by the project would be met by the Turkey's current national electricity grid. This would lead to 87,318 tCO₂e emissions per annum.

$$\begin{aligned}
 ER_y &= 87,318 - 0 - 0 \\
 &= 87,318 \text{ tCO}_2\text{e}
 \end{aligned}$$

2.5 Description of how the emissions of GHG by source in baseline scenario are reduced below those that would have occurred in the absence of the project activity (assessment and demonstration of additionality):

Referred by the Baseline Methodology, the "Tool for the Demonstration and Assessment of Additionality (Version 5.2)" outlines a step by step approach for the assessment of additionality or in other words the emission reductions that would have occurred in the absence of the project.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity

In the absence of the proposed project activity, plausible and credible project activities to the proposed project activity are as follows:

- Alternative 1 : Construction of a thermal (natural gas) power plant with the same annual power output
- Alternative 2 : The proposed project activity not undertaken as a VER project activity
- Alternative 3 : Construction of some other renewable energy plant with the same annual power output
- Alternative 4 : The same service of power supply is provided from the grid (Uzundere I HEPP is not built)

Although the coal and natural gas reserves in Turkey is very limited, there are many thermal power plants that have been being commissioned in recent years and Turkey's grid mainly consists of thermal power plants. However, lignite and coal fired power plants with the same annual output is not realistic alternatives, since these plants only become feasible with higher installed capacities. On the other hand, natural gas power plant with the same annual output is a realistic alternative. As they have become common practice in the energy sector due to off-take agreements provided by the government, these projects face easier circumstances in terms of financing and realisation. Also, the construction period of natural gas is lower than other alternative power plants.³⁷ Therefore, construction of a natural gas power plant with the same annual electricity output is a realistic alternative.

The Black Sea Region is not very rich of wind and the high forest density combined with very steep terrain does not allow for wind power plants. As it is seen in the REPA (Wind Energy Potential Map) of Rize, the areas which have enough potential for economically feasible wind farm investments are not very evident in the region.³⁸ As for the geothermal energy, this renewable energy source exists in west side of the country.³⁹ Therefore, at the project area, there is no geothermal source either. As a result, Alternative 3 for the region is eliminated.

As mentioned above, the Turkish Electricity Sector is heavily dependent on fossil fuel plants. According to a new bill that passed in 2001 (Electricity Market Law, Law No: 4628, Enactment Date: 20/02/2001)⁴⁰, the public funding in energy investments have stopped. The new bill allows incentives for private sector participation in energy investments. In addition to rehabilitation of existing plants, new capacities to be added to meet the increasing demand require mass amount of investments. Because the public funding of energy investments is no more legitimate, the new projects need to be accomplished by private sector. Yet, KARADENİZ cannot influence the future fuel mix in the energy market, but can only decide to invest on the project because KARADENİZ has no influence on the domestic energy policy of the decision makers.

Under these circumstances, the only realistic alternatives are 1, 2 and 4. Analyzing these three alternatives, it is seen that all scenarios are consistent with the baseline definition of ACM0002 / Version 11 where it defines the baseline scenario as the amount of electricity that would be delivered to the grid by the project activity, generated by the operation of existing grid-connected power plants and by the addition new generation sources, as reflected by the combined margin.

Sub-step 1b: Enforcement of applicable laws and regulations

The following applicable mandatory laws and regulations have been identified for the proposed project activity:

1. Electricity Market Law⁴¹
2. Law on Utilisation of Renewable Energy Resources for the Purpose of Generating Electrical Energy⁴²
3. Energy Efficiency Law⁴³

³⁷ <http://www.globalenerji.com.tr/hab-23000201-101,45@2300.html>

³⁸ <http://www.eie.gov.tr/duyurular/YEK/YEKrepa/RIZE-REPA.pdf>

³⁹ <http://web.deu.edu.tr/jenarum/index.php/turkyede-jeothermal>

⁴⁰ http://www.epdk.gov.tr/mevzuat/kanun/elektrik/elektrik_piyasalari_kanunu.pdf

⁴¹ Law number 4628, enactment date 03/03/2001, <http://www.epdk.gov.tr/english/regulations/electricity.htm>

⁴² Law number 5346, enactment date 18/05/2005, <http://www.eie.gov.tr/duyurular/YEK/LawonRenewableEnergyResources.pdf>

4. Forest Law⁴⁴
5. Environment Law⁴⁵
6. Regulation on procedures and principles of signing the agreement of utilisation of water resources for the purpose of electricity production in the electricity market⁴⁶
7. Regulation on Environmental Impact Assessment⁴⁷

Step 2: Investment Analysis

Sub-step 2a: Determine appropriate analysis method

The „Tool for the Demonstration and Assessment of Additionality’, version 05, from EB39, lists three possible analysis methods:

- Option I: Simple cost analysis;
- Option II: Investment comparison analysis; and
- Option III: Benchmark analysis.

Since the proposed project generates financial and economic benefits via the sales of electricity other than carbon revenues, Option I cannot be used.

Option II is only applicable to projects where alternatives should be similar investment projects in terms of generation capacity. Between Option II and Option III, benchmark financial analysis (Option III) is selected as the analysis method.

As a result Option III: Benchmark Analysis is selected as the analysis method and the Equity IRR is selected as the financial indicator for the demonstration of the additionality of the project.

Sub-step 2b: Option III. Apply Benchmark Analysis

The equity IRR is selected as the financial indicator. For the Benchmark Analysis, a relevant benchmark rate is calculated in line with the suggestion in “Tool for the demonstration and assessment of additionality” Version 05.2 which suggests to use the government bond rates, increased by a suitable risk premium. In relation to this, the benchmark rate is calculated in line with the Capital Asset Pricing Model (CAPM)⁴⁸. The CAPM suggests below formula to calculate the discount rate (in other words benchmark rate).

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$$

where:

- $E(R_i)$ is the expected return on the capital asset
- R_f is the risk-free rate of interest such as interest arising from government bonds
- β_i (the beta coefficient) is the sensitivity of the asset returns to market returns, or
also $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$,
- $E(R_m)$ is the expected return of the market

⁴³ Law number 5627, enactment date 02/05/2007,
http://www.eie.gov.tr/english/announcements/EV_kanunu/EnVer_kanunu_tercume_revize2707.doc

⁴⁴ Law number 6831, enactment date 31/08/1956. This will be made available to the DOE on request.

⁴⁵ Law number 2872. Published in Official Gazette No. 18132 on 11/08/’83. This will be made available to the DOE on request.

⁴⁶ National Gazette number 25150, 06/06/2003

⁴⁷ National Gazette number 26939, 17/07/2008

⁴⁸ Please see http://en.wikipedia.org/wiki/Capital_asset_pricing_model for the definition of CAPM

- $E(R_m) - R_f$ is sometimes known as the market premium or risk premium (the difference between the expected market rate of return and the risk-free rate of return).

Risk Free Rate (R_f)

The 30 year Eurobond which was issued by Treasury of Republic of Turkey was used as the proxy for the risk free rate in the country. This market return represents an after tax return over all components of the country's major index. The relevant interest at the date of investment decision of the facility which is assumed as the feasibility study of the facility completion date yield for this bond was 7.084% as of 30th of March 2007.⁴⁹

The Beta Coefficient (β_i)

The Beta coefficient of securities with values close to 1 determines the company securities as being of average risk. There are four power generating and trading companies under trade in the Istanbul Stock Exchange (ISE). The average Beta for these four companies can be calculated based the index created from these four stocks which is called as "Electricity Index" of ISE. The β_i for the Electricity Index is calculated as 0.922 by Bloomberg which is one of the prominent data suppliers in financial markets.

Market risk premium ($E(R_m) - R_f$)

To estimate the "Market risk premium" of Turkey, the independent study of Aswath Damodaran – Stern School of Business (New York University) was used. The study offers an assessment of the market risk for different countries according to their credit ratings as the determined market risk premium for Turkey is 9.29%, calculated on the basis of the credit rating Ba3 with positive perspective given to Turkey by Moody's.⁵⁰

All in all, the relevant discount rate is calculated as:

$$7.084\% + 9.29\% \times 0.922 = 15.65\%$$

The above discount rate can easily be assumed as conservative as the above benchmark assumes that the benchmark and the project have same natures in terms of risk. However, the Beta of the index includes lower risk compared to the project, as the index includes already established companies (which are up and running). However, it is obvious that the project includes more risk compared to already established companies as there are extra construction risks and inexperienced operational risks for the project. Therefore, the Beta of the project should be significantly higher than the index Beta. All in all, we can say that the CAPM derived benchmark rate is conservative and also in consistency with the benchmark rates received from local bankers mentioned above.

Sub-step 2c: Calculation and Comparison of Financial Indicators

Parameters required for calculation of key financial indicators

According to the financial feasibility study of the proposed project, the parameters required for the calculation of key financial indicators are in **Table 11**:

Parameter ⁵¹	Value
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⁴⁹ http://www.isyatirim.com.tr/report_download.aspx?file=4563

⁵⁰ <http://www.stern.nyu.edu/~adamodar/pc/archives/ctryprem07.xls>

⁵¹ Unless otherwise mentioned all figures are taken from the Financial Feasibility Study which was prepared in May 2010 by the company

Installed capacity (MW)	63.0
Grid connected output (MWh)	156,205
Capital investment (USD)	92,680,902
Income tax rate (%)	20.0
Expected tariff (USD cent/kWh) ⁵²	0.0749
Expected VER price (USD/tCO ₂ e)	5-7
Gen. Production Expenses (USD/year)	1,704,624
Project lifetime – (years)	35.0
Benchmark rate (%)	15.65

Table 11: Basic Parameters for Financial Analysis

IRR Calculation of the Project

IRR Calculation of the project is based on the feasibility study prepared for the Board of the Company in 2007. Aiming to show the effect of income from VERs for the financial performance of the project, two scenarios are presented in this section, one which excludes revenues from the sale of VERs and the second which includes the revenues from the sale of VERs. The assumptions used for this analysis are outlined as follows:

- The financial analysis is performed over the 35 year period from 2010 to 2045, this, therefore includes the investments made by the project owner between 2007 and 2010 and the operational costs along the lifetime of the project.
- The Equity IRR (Internal Rate of Return) of the project cash flow has been calculated.
- A tax rate of 20% is applied to the project in line with Turkish tax laws.
- The depreciation period of machinery and equipment was assumed as 30. years as the local law allows companies to depreciate machinery and equipment in 30 to 40 years (Please see http://www.alomaliye.com/2007/isa_yilmaz_amortismanlar.htm). In the meantime, the other fixed assets' (buildings etc) and unexpected costs' depreciation life are assumed as 40 years.
- The annual power generation figure is 156,205,000 kWh for the years between 2010 and 2019.
- The power purchase price for the project is assumed to be 7.49 US\$ cent per kWh which is calculated by converting 5.5 Euro cent using the monthly average exchange rate (15th August 2007-14th September 2007) before the date board decision regarding with investment dated 14th September 2007.
- The revenues from VERs are excluded from scenario 1, and included in scenario 2. The volume of VERs generated by the project is calculated by multiplying the annual electricity output of the project by the emission factor. For sensitivity analysis, the revenues related to the sale of the VERs are applied by multiplying the volume of VERs by the \$5, \$6 and \$7 price. The VERs are generated and sold between 2010 and 2019 for ten years.

Results

The cash flow analysis has been performed on two project scenarios;

- Scenario 1 – excludes revenues from the sale of VER
- Scenario 2 – includes revenues from the sale of VERs

As seen below, the project is not financially attractive in regular terms because the IRR of the project is estimated to be far less than benchmark expectations. Yet, the investor were appealed to subject project because of long term expectations such as a significant increase in energy prices due to increasing demand as well as an expectation of water flow increase and electricity production increase. Under regular

⁵² The equivalent of 5.5 Euro cent which was calculated by converting 5.5 Euro cent using the monthly average exchange rate (15th August 2007-14th September 2007) before the date board decision regarding with investment dated 14th September 2007

assumptions, contribution of carbon finance does not make the project leap over a benchmark that is acceptable to the banks. An IRR far less than the expectations of the potential financiers could provide additionality conditions and prove that the project is not attractive.

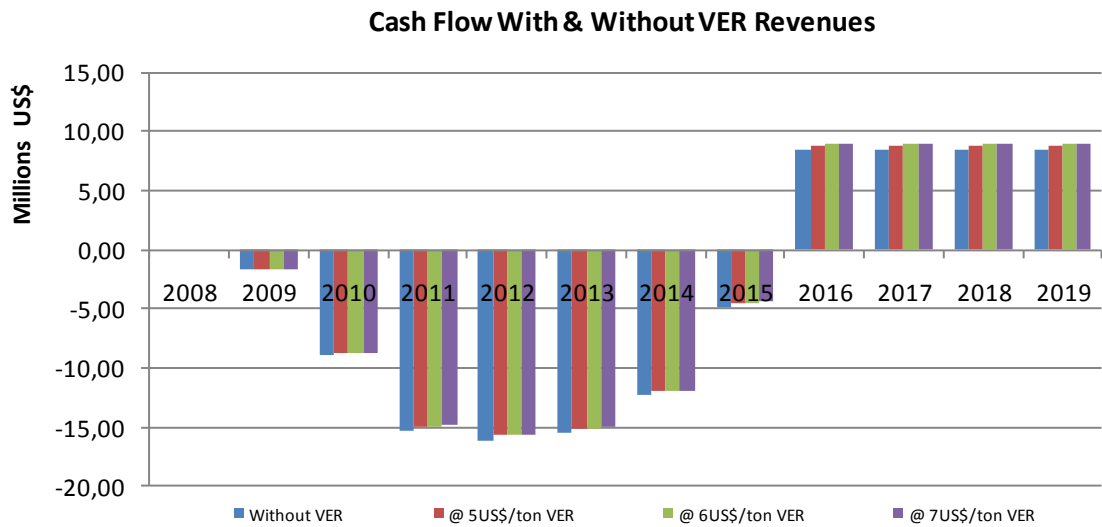


Chart 3: Project Cash Flow Comparison with and without VER

Below are the details of IRR estimations.

Sub-step 2d: Sensitivity Analysis

Sensitivity analysis was applied to Investment Costs, Energy Production and Electricity Sales Price. Applying variations on investment costs, the range was considered to be in a bandwidth of +10% and – 10% as advised in “Tool for the demonstration and assessment of additionality (Version 05.2).”

The sensitivity in the investment amount was applied to the investment items which represent equal or above 20% of the total investment amount as suggested by “Tool for the demonstration and assessment of additionality (Version 05.2).”

	IRR Sensitivity with Investment Costs				
Change in %	-10%	-5%	0%	5%	10%
Without VER	8,68%	8,42%	8,16%	7,92%	7,69%
IRR @5 US\$/ton VER	9,02%	8,74%	8,47%	8,22%	7,97%
IRR @6 US\$/ton VER	9,08%	8,80%	8,54%	8,28%	8,03%
IRR @7 US\$/ton VER	9,15%	8,87%	8,60%	8,34%	8,09%

Table 13: Sensitivity to Variation in Investment Costs

Variation on energy production was also applied in a bandwidth of +10% and –10% on the secondary energy production which reflects the volatility in calculations. Please note that “primary” part of the production reflects the electricity production which is 90% guaranteed amount. Therefore no sensitivity was applied to this part.

	IRR Sensitivity with Electricity Production				
Change in %	-10%	-5%	0%	5%	10%
Without VER	6,72%	7,44%	8,16%	8,89%	9,63%
IRR @5 US\$/ton VER	6,97%	7,72%	8,47%	9,23%	10,01%
IRR @6 US\$/ton VER	7,03%	7,78%	8,54%	9,30%	10,09%
IRR @7 US\$/ton VER	7,08%	7,83%	8,60%	9,38%	10,17%

Table 14: Sensitivity to Variation in Energy Production

Due to the fact that the feed in guaranteed tariff prices are fixed in Turkey, the unit energy price is not considered to be an exogenous variable. However, sensitivity on the upward side was also applied to see the effect of price changes.

	IRR Sensitivity with Electricity Price				
Change in %	-10%	-5%	0%	5%	10%
Without VER	na	na	8,16%	8,89%	9,63%
IRR @5 US\$/ton VER	na	na	8,47%	9,22%	9,98%
IRR @6 US\$/ton VER	na	na	8,54%	9,28%	10,05%
IRR @7 US\$/ton VER	na	na	8,60%	9,35%	10,12%

Table 15: Sensitivity to Variation in Energy Production

The above sensitivity analysis reveals the fact that no alternative scenario without VER revenues can make the project pass the benchmark IRR expectation which is 15.65% . Therefore, the project is not financially attractive.

Step 4: Common Practice Analysis

Sub-step 4a: Other activities similar to the proposed project activity

Analyzing similar activities in the region that means the same geographical region where the subject project is located, it is seen that there are 7 more hydro-power plant projects which are similar to the subject project in installed capacity of ranging between 53 MW and 73 MW, -10 MW and +10 MW of the installed capacity of Uzundere 1 HEPP in Eastern Black Sea Region (Among Artvin, Giresun, Gumushane, Ordu, Rize and Trabzon provinces).

No	Company Name	Project	Location	Type	Installed Capacity (MWm)	Electricity generation (kWh/year)	Completion Rate - as of May 2010 (%)
1	AES-İC İtaş Enerji Üretim ve Ticaret A. Ş.	Güneyce Dam and HEPP	Trabzon	Hydro-Dam	64.6	301,810,000	0.1
2	Ebara Elektromekanik Enerji San. Ve Tic. Ltd. Şti.	Cüneyt Weir and HEPP	Artvin	Hydro-Runoff River	60.7	189,020,000	3.2
3	Erarı Elektromekanik Enerji San. Ve Tic. Ltd. Şti.	Damla Weir and HEPP	Artvin	Hydro-Runoff River	56.0	182,313,000	3.2
4	İdil İki Enerji San ve Tic A.Ş.	Kozbükü HEPP	Ordu	Hydro-Runoff River	67.8	238,582,000	1.2
5	Karhes Karadeniz Hidroelektrik Enerjiden Elektrik Üretim Santralı Ltd. Şti.	Çırakdamı HEPP	Giresun	Hydro-Runoff River	58.7	140,000,000	85.1
6	Karhes Karadeniz Hidroelektrik Enerjiden Elektrik Üretim Santralı Ltd. Şti.	Dereli HEPP	Giresun	Hydro-Runoff River	58.8	157,500,000	72.9
7	Suata Enerji Müt. Müş. San. ve Tic. Ltd.Şti.	Burçak I-II Weir and HEPP	Giresun	Hydro-Runoff River	72.5	223,870,000	46.1

Table 16: List of privately held HEPP projects, similar to the project activity, under construction in Eastern Black Sea Region⁵³

All these projects are run-off-river type hydro projects (except Güneyce HEPP which is a reservoir type hydropower project) and private sector investments. Also, the completion rates of these projects are too low, when it is compared to the subject project of which completion rate is 100%. On the other hand, it is not possible to compare all these projects with the subject project in respect to carbon revenues. There is no detailed information accessible for the similar projects if they benefit from carbon revenues or not, since there is no registration authority in the host country to apply for this information.

As also mentioned above, participation of private sector in the electricity generation from hydropower plants is a very new concept in Turkey. Facing the growing demand for electricity and lacking the capital to realize hydro investment, the State outsourced the construction of those plants through licenses. As seen in the table below⁵⁴, the number of projects that are close to completion is very low (only 57 projects out of 477 were completed more than 50% as of now).

⁵³ http://www.epdk.gov.tr/lisans/elektrik/ilerleme_proje.htm (Accesses on 22 November 2010)

⁵⁴ http://www.epdk.gov.tr/lisans/elektrik/ilerleme_proje.htm (Accesses on 22 November 2010)

Completion Ratio (%)	Number of Facilities
>10% Completed	143
>20% Completed	108
>30% Completed	92
>40% Completed	70
>50% Completed	57
>60% Completed	40
>70% Completed	34

Table 17: Number of HEPP facilities completed over a certain completion ratio

In addition, the new capacity from privately owned hydro facilities represents a fraction of new facilities added to Turkey's capacity every year. As displayed in the table below, on average, the new privately owned hydro facilities represents only 2.3% of new additions. Similarly, new additions from privately held hydro projects represents only 0,1% of Turkey's installed capacity in the last 6 years.

	2007	2006	2005	2004	2003	2002
Total Additional Installed Capacity from Private Hydro* (MW)	36,1	105,4	45,6	9,5	36,6	28,0
Total Additional Installed Capacity (MW)	1.059,6	1.817,6	2.019,5	1.236,9	3.741,3	3.513,4
Turkey's Total Installed Capacity	40.835,7	40.564,8	38.843,5	36.824,0	35.587,0	31.845,8
Private Hydro / Total Addition	3,4%	5,8%	2,3%	0,8%	1,0%	0,8%
Private Hydro / Total Capacity	0,1%	0,3%	0,1%	0,0%	0,1%	0,1%

*: Excluding Concessionary Rights and BOT Projects

Source:

<http://www.teias.gov.tr/ist2007/8.xls>

<http://www.teias.gov.tr/ist2006/8.xls>

<http://www.teias.gov.tr/istatistik2005/7.xls>

<http://www.teias.gov.tr/istat2004/7.xls>

<http://www.teias.gov.tr/istatistik/7.xls>

Table 18: Annual privately held hydro projects' capacity additions, total additions and total installed capacity.

What is more, the rate of electricity generation of these privately held HEPPs are much less than the rate of installed capacities of these projects. The additional electricity generation from private hydro projects are only 3% of the total additional capacity and only, 0.1% of Turkey's total electricity generation in 2007. Also, the total electricity production of private hydro projects has a rate of 2.5% in Turkey's total electricity generation.

Sub-step 4b: Discuss any similar Options that are occurring

There is no accessible information on the projects benefit from carbon revenues. Therefore, it is not possible to compare the projects with the subject project in respect to carbon revenues. However, the low rate of completion of the projects and the low contribution privately held hydro projects confirm and proves that the electricity generation from HEPP business is not a common practice. Therefore Step 4 is satisfied.

3 Monitoring:

3.1 Title and reference of the VCS methodology (which includes the monitoring requirements) applied to the project activity and explanation of methodology choices:

The consolidated monitoring methodology for grid connected generation from renewable sources named as “Approved Monitoring Methodology ACM0002 / Version 11 is applied. The monitoring methodology is applicable as the data will be archived electronically and be kept at least for 2 years after the end of the last crediting period. All measurements will be conducted with calibrated measurement equipment according to relevant industry standards.

The application of ACM0002 / Version 11 to Uzundere I HEPP is justified because;

- Uzundere I 63 MW Runoff-River HEPP Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources and which supplies electricity to a system that is supplied by at least one fossil fuel fired generating unit,
- The project does not involve switching from fossil fuels to renewable energy at the site of the project activity,
- The project activity is not implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m².
- Since the project activity involves a very small reservoir, it does not result in a power density of lower than 4 W/m², as per definitions given in the Project Emissions section.
- 5 years of historical data are available.
- The project involves construction of new units in a brand new plant, in other words the project does neither involve the addition of renewable energy generation units at an existing renewable power generation facility nor does it foresee to retrofit or modify an expired facility of renewable energy generation.

3.2 Monitoring, including estimation, modelling, measurement or calculation approaches:

This monitoring plan is prepared as a road map for the verification of the information which needs to be monitored periodically.

All relevant baseline emission factors are defined ex-ante. Please see the baseline estimations –Section 2.4- for Operating and Build Margin estimations. Hence, there are three sets of data to be monitored during the project activity.

The first one is the amount of electricity fed into the grid. There is already an electricity meter installation plan to provide the data that is required by the monthly invoicing process to TEIAS. Uzundere I HEPP is a new power unit which did not replace a fossil fuel plant on the project site. The plant manager of Uzundere I HEPP will be responsible for providing the electricity meter readings which shows the electricity generation of the company.

In the powerhouse, there will be two electricity meters in order to measure the produced electricity accurately. One is the main meter for measuring and the other is the check meter for control. Both of these meters are metering the energy in two directions (consumption and production). If there is a measuring difference between these two meters and one of the parties (TEIAS or the company) requests for calibration of the meters, in this case, the meters will be calibrated without waiting for the periodic calibration date. (TEIAS System Usage Agreement, Art 3, B./2./b))⁵⁵ This calibration process is done by

⁵⁵ www.teias.gov.tr/sistemkullanim1.doc

another third party under the control of TEIAS. The company is not responsible for calibration of the meters in Turkey according to the local standards.

The second data to be monitored is the installed capacity of the project activity. The data will be monitored from the supplier information on the equipments which shows the installed capacity of the turbines. Since the installed capacity is not a dynamic value, there won't be any continuous measurement as it is done in electricity generation measurement. The plant manager will be responsible to provide the installed capacity data.

The third data to be monitored is the reservoir area. This data will be calculated from coordinate readings which are gathered through topographical surveys or GPS readings. The data will be collected by the project owner during the period when the water flow is high.

The data will be collected and stored by KARADENİZ not only during the crediting period but also two years after the last issuance of VCUs to Uzundere I HEPP project for that crediting period.

The emission factors are calculated ex-ante for the period of ten years. Thus no monitoring and recalculation of the emissions factor during the crediting period is required as suggested in the *Methodological Tool to calculate the emission factor for an electricity system*. Please note that the project is a zero emission generating activity and no particularly sensitive or critical sustainable development indicator was identified.

ID Number	Data Variable	Source of Data	Data Unit	Measured (m) Calculated (c) Estimated (e)	Recording Frequency	Proportion of data to be monitored	How will the data be archived? (electronic or paper)	Comment
1. EG _{facility,y}	Quantity of net electricity delivered to the grid.	Metering device from the grid operator	MWh	m	Continuously	100%	Electronic and paper	The data will be obtained from the monthly meter readings. "Meter reading document" is already required for the invoicing to TEIAS.
2. Cap _{PJ}	Installed capacity of project activity after implementation	Project site	W	m	Yearly	100%	Electronic and paper	
3. AP _J	Area of the reservoir measured in the surface of the water, after implementation of project activity, when reservoir is full	Project site	m ²	c	Yearly	100%	Electronic and paper	Data will be obtained from the yearly readings
4. EF _{grid,CM,y}	Combined margin CO ₂ emission factor for grid connected power generation in year y	As per the "Tool to calculate the emission factor for an electricity system"	tCO ₂ / MWh	c	Calculated once for each crediting period	100%	Electronic and paper	Calculated once for each crediting period
5. FC _{i,y}	Amount of fossil fuel type i consumed in the project electricity system	As per the "Tool to calculate the emission factor for an electricity system"	Ton (m ³ for Natural Gas)	c	Calculated once for each crediting period	100%	Electronic and paper	Calculated once for each crediting period
6. NCV _{i,y}	Net calorific value of Fossil Fuel Type i in year y	As per the "Tool to calculate the emission factor for an electricity system"	TJ/Ton (TJ/m ³ for Natural Gas)	c	Calculated once for each crediting period	100%	Electronic and paper	Calculated once for each crediting period

Table 19: Data to be monitored and monitoring process

3.3 Data and parameters monitored / Selecting relevant GHG sources, sinks and reservoirs for monitoring or estimating GHG emissions and removals:

3.4

Data Parameter	EGfacility,y
Data Unit	MWh
Description	Quantity of net electricity delivered to the grid
Source of Data	Project site
Measurement Procedures (if any)	MWh
Value applied	156,205 MWh
Monitoring Frequency	Continuously
QA / QC Procedures	Double check with the grid operators
Any comment	N/A

Data Parameter	CapPJ
Data Unit	MW
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of Data	Project site
Measurement Procedures (if any)	Supplier information on the equipment
Value applied	63 MW
Monitoring Frequency	Yearly
QA / QC Procedures	Supplier information on the related equipment and existence of the equipment will be checked
Any comment	N/A

Data Parameter	APJ
Data Unit	m2
Description	surface of the water, after the implementation of the project activity, when the reservoir is full
Source of Data	Project site
Measurement Procedures (if any)	Topographical surveys, GPS readings
Value applied	7,275.54 m2
Monitoring Frequency	Yearly
QA / QC Procedures	The readings will be done during the period when the water flow is high to have the largest reservoir area
Any comment	N/A

Table 20: Data to be monitored and parameters

3.5 Description of the monitoring plan

Data	Uncertainty Level of Data (High/Medium/Low)	Explain QA/QC Procedures Planned for this Data or why such procedures are not necessary
Quantity of net electricity delivered to the grid	Low	The metering equipment will be properly calibrated and checked periodically for accuracy, to ensure that any error resulting from such equipment shall not exceed +0.2% of full scale rating. Rules for meter accuracy are stated in the TEIAS connection agreement. In addition to the main meter, a redundant meter will be installed to exclude the possibility of data loss.
Installed capacity of the project after the implementation	Low	The installed capacity of the project is equal to the total installed capacities of the equipment implemented at the beginning of the project's life. The installed capacity will be monitored from supplier's documents related to the equipment.
Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full	Low	The measuring will be done through either topographical survey or GPS readings. In either ways, sample coordinates from the reservoir border will be read. Then, these data will be used to calculate the reservoir surface area.

Table 21: Quality Control of Monitoring

The first data that should be monitored is the amount of electricity ($EG_{\text{facility},y}$) that is annually fed into the grid by the plant. The project owner, Karadeniz Elektrik Uretim Sanayi A.Ş., has already signed an agreement with TEIAS that includes measuring the amount of electricity that goes to the system accurately. Based on this sales agreement, there is no need for an additional process or equipment to monitor the data.

The second data to be monitored is the installed capacity of the project activity. The data will be monitored from the supplier information on the equipments which shows the installed capacity of the turbines.

The third data to be monitored is the reservoir area. The reservoir area is expected to be topped when the water flow is peaked. The area can be calculated by making coordinate readings around the border of the reservoir. The coordinate readings will be done when the water flow is at its highest level.

Each monitoring period is planned to last one year. At the end of each period, the reservoir area will be calculated, installed capacity will be read from the digital displays and the annual net electricity generation will be calculated as an accumulation of electricity meter readings over the year. The total amount of the net electricity generation of that period will then be multiplied with the combined emission factor. An excel sheet that also includes the estimation of the combined emission factor will be used to calculate the amount of ERs through the multiplication of net electricity generated and combined emission factor. Thus, a transparent and traceable baseline approach and ER calculation will be provided. For quality assurance and follow-up, GAIA Carbon Finance will be present at each monitoring assembly. The PMUM records will be cross-checked with the monthly electricity meterings as well.

For each crediting period, the EF value will be recalculated as per the latest version of the “Tool to calculate the emission factor of an electricity system” at that date.

The management of KARADENİZ is responsible with all monitoring and reporting issues in general.

Name of person/entity determining the monitoring methodology

GAIA Finansal Danismanlik, Turkey
Tel: 0090 (0) 212 224 0450

Fax: 0090 (0) 212 224 0466

Email: info@gaiaacf.com

4 GHG Emission Reductions:

4.1 Explanation of methodological choice:

The baseline for the project was established through the official methodology of ACM0002 / Version 11, named “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” and “Tool to calculate the emission factor for an electricity system, Version 02” as approved by the CDM Executive Board. Conservative options and data were selected during the implementation of the methodology.

4.2 Quantifying GHG emissions and/or removals for the baseline scenario:

Please see section 2.4.

4.3 Quantifying GHG emissions and/or removals for the project:

Since the project is a run-off river hydro scheme and the reservoir area of the facility is very small in comparison with the installed capacity of the project, PD(Power Density) is much more than 10 W/m². Thus, the project emissions (PEy) are assumed to be 0 in line with ACM0002 / Version 11 “Project emissions: For most renewable energy project activities, PEy = 0. However, for following categories of project activities, project emissions have to be considered.”

Anthropogenic emission reduction is a function of the expected net amount of electricity produced, for the given project, which amounts to 156,205 MWh per annum. The annual baseline emissions amount to 87,318 tCO₂e. Over the period of ten years, the total amount of baseline emissions will reach 873,180 tCO₂e. For all calculations, the rounded value of the combined emission factor, 0.559 is applied.

4.4 Quantifying GHG emission reductions and removal enhancements for the GHG project:

Please see section 2.4.

5 Environmental Impact:

There is no endangered fauna and flora in the region. The project design preferred CTP pipes rather than open canals where necessary to minimize the project impact on local households.

The project has no trans-boundary environmental and social impacts.

The domestic law and regulation does not require any environmental impact assessment. According to applicable law, the project owner is required to conduct a pre-assessment where they need to prove that the project site is not particularly sensitive to any environmental and social impact. Such pre-assessment has been conducted and the project has been qualified not to have any further environmental and social assessment. (Please see Annex 4)

The project design identifies and addresses all social and environmental issues. The project pursues a pre-assessment report approved by the Ministry of Environment where all environmental and social concerns were identified through non-formal meetings and hearings with local stakeholders. The pre-assessment report asserted that all issues were taken into consideration in the project design. The related law named as

Annex V – List of Sensitive Areas list the aquifers in the country that require special attention. The project paid attention to location of the listed aquifers and assured that the project design does not harm any aquifer in the region.

As mentioned below in Section 7, the first feasibility study for the project which was designed for 43 MW of installed power by Karadeniz was originally finalized in June 2005. In 2006, the license was granted for that first feasibility study. Contemporarily, it was figured out that it was possible to realize a new design for a 63 MW of installed power facility after detailed review of the water flow data and the new feasibility studies was started. Early 2008, the application for that increased installed power was made with a new feasibility study. Revised license was granted at late 2009.

For both design the EIA Exemption Letters were granted as can be seen in Annex 4. But, later reports were cancelled by Rize Administrative Court and the Court has cancelled the both of Exemption Letters, and ordered the preparation of an EIA. Upon this order, an EIA Report was prepared in 2010 and extended to the official authorities, and a Positive EIA decision document dated 1st April 2010 was granted as can be seen in Annex 5. The stakeholder meeting for the preparation for the Report was held in 14th April 2009.

The headlines of the EIA Report prepared in 2010 are as follows:

Karadeniz Elektrik Üretim Sanayi A.Ş. is planning to construct and operate Uzundere-I 63 MW HEPP near Cataldere village, in borders of Cayeli district of Rize province in Turkey.

Within the scope of the EIA of the project, a Stakeholder Participation Meeting was held on 14.04.2009 at 13.00 in Cukurluhoca Village Coffeehouse, Cayeli, Rize.

Approaching roads, power house, conveyance tunnels, approaching tunnels and Incesu weir, penstock, switchyard and loading gate construction works were almost finished. After AIA process is completed, Uzundere-I weir construction works to be finished and the project will be commenced in a 1 year period. Most of the project area is forestry lands and the company obtained all necessary permits for the use of these lands.

From both weirs, Incesu and Uzundere weirs, at least 330 lt/sec water will be released regularly. At project location, there is no irrigation activity and this water will be enough for natural river life. As the evaporation is very low, rainfall is high and there are many lateral rivers in the region, the water flow will be much higher than the water released from the weirs.

The only project item whose excavation activities has not started is Uzundere-I weir. There will be 40.000 m³ excavated earth in Uzundere-I weir construction and 800 m³ of this earth will be vegetable soil. This vegetable soil will be stored appropriately and used for landscaping purposes after the construction. Some of the excavated earth will be used as filling material at construction of the weir and the rest will be stored at storage areas.

The possible dust emission is calculated for the project and the results show that the emission amount is under the allowed limits in the “Regulation on Control of Industrial Air Pollution (dated 03.07.2009, numbered: 27277)”. In order to protect the air quality in the region, the construction activities will be executed in line with the related laws and regulations.

Domestic, solid and liquid wastes will be disposed in line with the regulations.

The limited damages to the environment arising from the project during the construction will be minimized by taking all necessary precautions. Also, in order to protect riverine habitats and Salmo Labrax species, the project will continue by taking necessary precautions.

The company made commitments to the Ministry of Environment and Forestry to minimize environmental impacts of the project. In addition to the above explanations, some of the commitments made by the company are as listed below:

- No waste will be thrown to the environment. The wastes will be disposed in line with the regulations.
- The excavated earth will be stored in line with the regulations and the construction site will be sprinkled to decrease dust emissions.
- Sands and gravels will be provided from the region.
- There will not be any folding or grinding activities at construction site. Ready mixed concrete will be used in the construction of the project items. Therefore, there won't be dust emissions rising from these activities.
- There will be an infirmary at project site.
- The noise rising from the construction and operation of the plant will not exceed the allowed limits. The workers will work 8 hours a day and this will not have any impact on occupational health.

6 Stakeholders' comments:

Although the domestic laws and regulations (Regulation on Environmental Impact Assessment (Date: 16.12.2003, Number: 25318), and Regulation on Environmental Impact Assessment (Date: 17.07.2008, Number: 26939). According to the Article 7 of both do not require any stakeholders' participation, both in the project planning and construction phase, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings.

In these meetings authorised people from KARADENİZ introduced project to the local people in terms of the size of the project, expected length of the construction period and to give details about how this project will impact their lives. The information given mostly focused on how the project might have some environmental effects and how these issues will be mitigated by the investor. These meetings were finalized in November 2007 (These meetings were the ones held before the Rize Administrative Court and the Court has ordered the preparation of an EIA. The summary of that report was given above in Section 5-Environmental Impact.).

The feedback from the stakeholders was reflected on the project design and implementation. Also, a contact person from KARADENİZ and relevant contact information was announced in the relevant meetings for ongoing communication.

The project owner used different channels to invite comments by stakeholders. Most of the meetings were held through the Mukhtar (village governors) and Council of Elderly (İhtiyar Heyeti; the village council) of Cataldere village, and the officials of Municipality of Kaptanpasa. The project owner visited various households in the project area to comprehend their needs and to identify how the project could assist in meeting the social and economic needs of the community.

Below is the list of stakeholder comments and the details of those comments were taken into account.

- The stakeholders (Mukhtar Mustafa Karakas) addressed unemployment and weak economic activity as the major social problems in the area.

Within the context of the project 120 workers were employed. The local workers will be given priority during the hiring for plant operation. Also, during the construction, construction machines and equipment have been hired from the vicinity of the project site. The project owner has procured the all needs of the construction site from local resources in order to contribute to the local economy of the region.

➤The stakeholders (Naci Ayar and Hüseyin Turan)(asked for assistance for some social needs of Cataldere Village

The project owner helped local people and the institutions in many ways, from renewing the electricity infrastructure to building roads in the forest to help forest workers in fighting against possible fires, as it is listed under the section 1.16.

The second round of stakeholder consultation meeting within the scope of EIA which was ordered to be prepared by Rize Administrative Court was held in 14th April 2009. The attendants repeated the same comments that they did in previous meetings as briefed above in this section. The headlines of the mentioned EIA Report was briefed in Section 5.

7 Schedule:

The first feasibility study for the project which was designed for 43 MW of installed power by Karadeniz was originally finalized in June 2005. After that, the detailed project design work was completed and search for financing of the project started in September 2006. In 2006, the license was granted for that first feasibility study. Contemporarily, it was figured out that it was possible to realize a new design for a 63 MW of installed power facility after detailed review of the water flow data and the new feasibility studies was started . Early 2007, the first civil works were started for the project. The same year, it was decided to increase the installed power to 63 MW by the board of KARADENİZ. Early 2008, the application for that increased installed power was made with a new feasibility study. Revised license was granted at late 2009.

During the search of financing, the project owner was aware of that GHG emission trade could be an option to straighten the feasibility of the project. The schedule below displays various stages of project activity.

Project Activity	Q205	Q305	Q405	Q106	Q206	Q306	Q406	Q107	Q207	Q307	Q407	Q108	Q208	Q308	Q408	Q109	Q209	Q309	Q409	Q110	Q210	Q310
Feasibility study completed																						
Design completed																						
GHG emission reduction opportunity identified																						
Applications submitted to the Banks																						
Banks approve debt facility																						
Civil works commenced at site																						
Commissioning/testing of HEPP																						
Operation, delivery of power to the national grid.																						

Table 22: KARADENİZ, Uzundere I HEPP Project Schedule

The license duration of the project is 49 years starting from August 31st, 2006 and the first crediting period is 10 years starting from May 27th, 2010. The crediting period will be renewed two more times according to the approved methodology.

During the crediting period, the defined Monitoring Plan will be executed for each year and the realised VERs will be verified by a DOE.

8 Ownership:

8.1 Proof of Title:

Please see Annex 2 for licence.

8.2 Projects that reduce GHG emissions from activities that participate in an emissions trading program (if applicable):

Not applicable.

ANNEX 1 QUALITY OF REDUCTIONS

Compliance with the National Environmental Standards

The Uzundere I HEPP is located near Catal Creek. The project is located within the province of Rize and it is situated in a valley. It will be constructed with the aim of utilisation of river for electric energy production. According to the current Turkish legislation, the construction, operation and location of a Uzundere I HEPP should conform to the environmental protection law and standards. The laws and ordinances to which it should correspond are:

Turkey's environmental legislation⁵⁶

The Environmental Law No. 28721 is the primary legislation governing environmental protection and improvement of the environment, appropriate and efficient use and protection of natural resources and land, the prevention of pollution and the protection and maintenance of wildlife and plantations. On 26 April 2006, Law No. 5491 was passed; revising, modernizing and expanding the Environmental Law ("**Revised Environmental Law**"). While keeping the general outline and spirit of the original law, the Revised Environmental Law is more specific on various issues and embraces a more modern perspective regarding environmental issues, with a view to draw level with EU legislation.

Core Principles and Purpose

The Revised Environmental Law introduces two principles as the main purpose of this law, namely a sustainable environment and sustainable development. The sustainable environment principle is defined as the *"improvement, protection, and development of all environmental assets that constitute the environment of the present and*

⁵⁶ Published in the Official Gazette dated 11 August 1983 and numbered 18132.

future generation in every (social, economical, physical, etc.) way without endangering the existence and quality of the resources that will be needed by future generations." Similarly, the sustainable development principle is defined as the *"development and growth based on the principle of establishing balance between the environmental, economical, and social targets, which ensure that future generations shall live in a healthy environment."*

The expansion of the Revised Environmental Law's definition section is noteworthy, not only due to its inclusion of these two core principles, but also due to its definitive coverage of various concepts and entities new to the Turkish environmental legislation, such as 'biological diversity', 'ecological balance', 'sensitive lands', 'environmental management' and 'non-ionizing radiation'.

The High Council and the Ministry

The High Council of Environment, members of which shall be appointed by the Prime Minister from amongst the Council of Ministers and shall be chaired by him, has replaced the former Central Environmental Council. The High Council of Environment is responsible for, and authorized to, among others, (i) determining the targets, policies and strategies to be implemented to ensure an efficient environmental management; (ii) adopting legal and administrative measures that enable environmental issues to be considered with respect to economical decisions under the sustainable development principle; and (iii) resolving disputes on environmental issues concerning different ministries. The Ministry of Environment and Forestry ("**Ministry**") is the sole body authorized to inspect compliance with the provisions of the New Environmental Law. The Ministry may delegate this duty to the relevant City Private Administration, municipalities that established environmental management units, and certain other administrative bodies. The Ministry is also in charge of implementing all types of legal, technical, fiscal, and administrative measures to comply with national obligations that arise due to ratification of international treaties on the environment.

Protection of the Environment

Turkish law prohibits the direct or indirect release, storage, transportation and dislodgement of any type of waste or residue in a manner that may harm the environment. Entities falling within the scope of the relevant prohibitions, are obligated to take preventive measures if there is a possibility of pollution; to stop pollution; and to lessen or restore its impact on the environment if pollution has taken place. The Ministry shall procure and implement country-wide zoning plans on the scale of 1/50,000 and 1/100,000 to prevent environmental pollution that may arise due to meeting various needs of urban and rural populations.

The protection of biological diversity and ecological balance are essential. Commercial activities regarding endangered fauna and flora as well as rare species, unless otherwise allowed by legislation, are prohibited. Similarly, marine fish farms cannot be established either in any closed coves bearing a sensitive-lands quality or within natural and archeologically protected areas. All types of resources, income, fees and fines that are collected and gained within the framework of environmental legislation shall be allotted towards environmental matters before all else.

Hazardous Chemicals and Wastes

Importing hazardous wastes into Turkey is strictly prohibited by the Revised Environmental Law and stringent provisions regarding the handling, production, storage, transportation, annihilation of hazardous chemicals and wastes are also introduced. Entities that produce, sell, store, use or transport hazardous chemicals; or those that collect, transport, store, recycle, re-use or annihilate hazardous wastes are severally liable in respect of various obligations for different types of wastes, as detailed in the Revised Environmental Law. For example, such entities must procure *"hazardous chemicals and wastes financial obligation insurance"* and obtain the Ministry's permission before commencing their operations.

Shutting Down Operations, New Administrative Fines, Criminal Behaviours

One of the most significant aspects of the Revised Environmental Law is the strict application with regard to doling out penalties and fines, which previously were often deserved but rarely applied. The Revised Environmental Law provides that, should the activities of any persons or entities be found to be in

violation of the applicable environmental regulations, the Ministry⁵⁷, in its sole discretion, may grant such persons or entities a one-time-only grace period of less than one year. When a grace period has not been granted, the operations of such violating-party shall be immediately shut down; whereas when a grace period has been granted, operations shall be shut down if such violation has not been rectified at the end of such period.

Another equally important aspect of the Revised Environmental Law is the significant increase of administrative fines. For instance, those who produce, sell, store, use or transport hazardous chemicals or who collect, transport, store, recycle, re-use or annihilate hazardous wastes in violation of the relevant legislation, are subject to administrative fines ranging between 100.000 YTL and 1.000.000 YTL (approximately €75,000 and €500,000, respectively).

A penalty of between six months and one year of imprisonment may be imposed on those who provide wrong or misleading information regarding environmental matters, in violation of the obligation to notify and provide information. The provisions of the (New) Turkish Criminal Code regarding "counterfeiting documents" shall be applicable to those who issue false or misleading documents in matters regarding the application of the Revised Environmental Law. Further, the provisions of the (New) Turkish Criminal Code, regarding intentional and negligent pollution of the environment, imposing monetary as well as imprisonment penalties, shall come into effect on 12 October 2006.

Financial Incentives

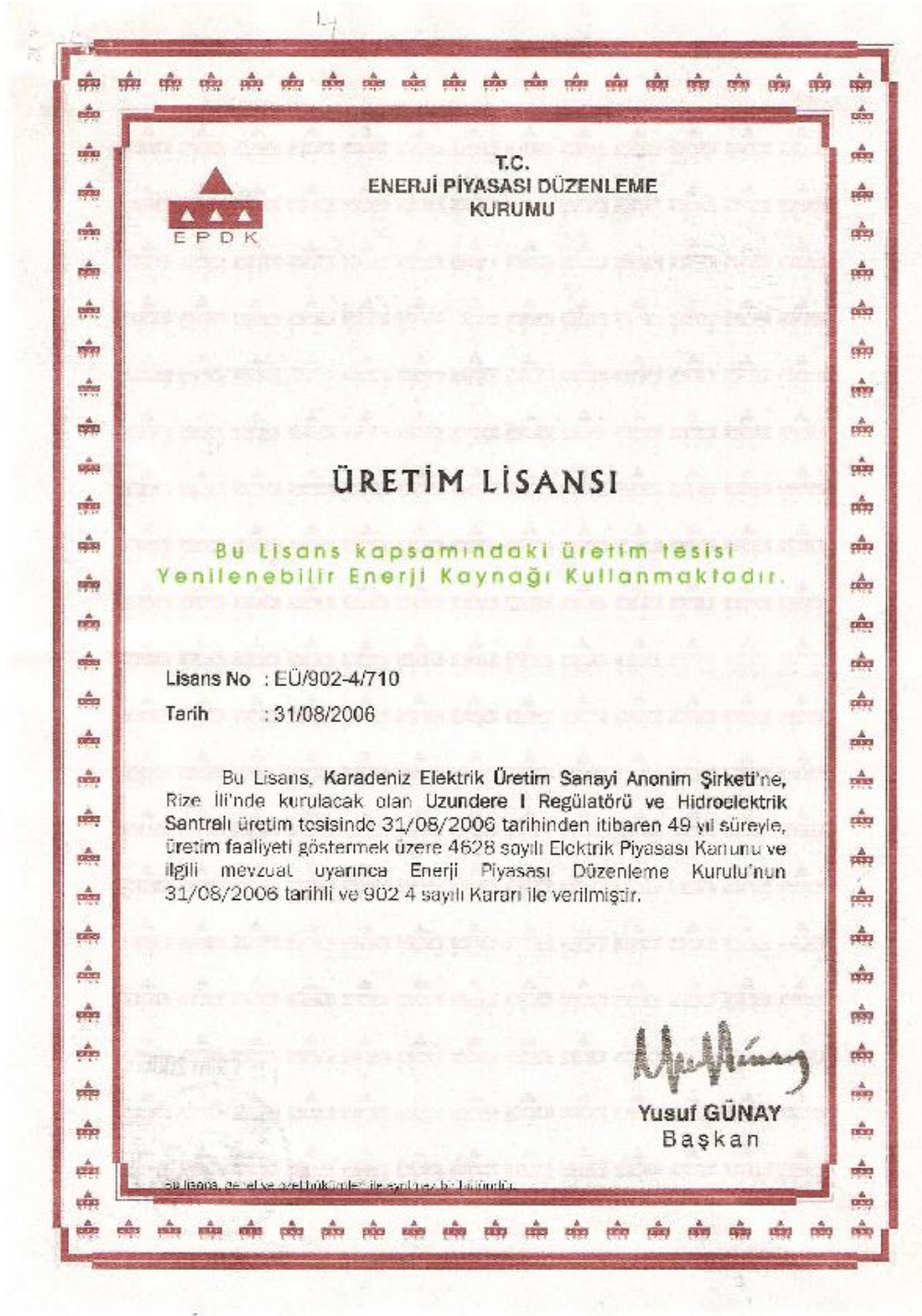
The Revised Environmental Law sets forth financial and fiscal incentives for activities focused upon the prevention and restoration of environmental pollution. Persons and entities that establish and operate treatment facilities and meet their various obligations under relevant regulations may receive a credit of up to 50% of their energy costs.

ANNEX 2

Proof of Title

The document below explains that this license is provided to Karadeniz Elektrik Uretim Sanayi A.S. for Uzundere I project which is planned to be installed in Rize province, in accordance with Energy Market Law and the Decision of Energy Market Regulatory Authority (Date: 31.08.2006, Number: 902-4)

⁵⁷ Or the authorized bodies to whom the Ministry's inspection authority has been delegated.



ANNEX 3

Details of OM Calculation

Gross/Net Electricity Generation

VCS Project Description Template

Electricity generation	2002	2003	2004	2005	2006	2007	2008
Gross Electricity Production	129.399,5	140.580,5	150.698,3	161.956,2	176.299,8	191.558,1	198.41
Net Electricity Production	123.726,8	135.248,3	145.065,7	155.469,1	169.543,1	183.339,7	189.76
Net/Gross	0,956	0,962	0,963	0,960	0,962	0,957	0,956
Net Electricity Production from Thermal Sources	91.207,7	101.003,0	100.459,1	117.228,4	126.634,4	148.333,3	156.76

IPCC Values

	kg CO ₂ /GJ			Default Carbon Oxidation Factor
	min	mid	max	
hard coal	92,8	96,1	100,0	1,0
lignite	90,9	101,0	115,0	1,0
fuel oil	75,5	77,4	78,8	1,0
diesel oil	72,6	74,1	74,8	1,0
natural gas	54,3	56,1	58,3	1,0
lpg	61,6	63,1	65,6	1,0
naphtha	69,3	73,3	76,3	1,0

Source: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 Energy, Chapter 1 Introduction, Table 1.4
http://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

Fuel Consumption

	Fuel Consumption (FC i,y) [Ton]						
	2002	2003	2004	2005	2006	2007	2008
hard coal	2.050.700	3.705.686	4.564.713	5.259.058	5.617.863	6.029.143	6.270.008
lignite	42.576.239	35.556.028	33.776.660	48.319.143	50.583.810	61.223.821	66.374.120
fuel oil	3.180.701	2.864.392	2.403.338	2.005.899	1.746.370	2.250.686	2.173.371
diesel oil	98.374	14.123	29.141	28.442	61.501	50.233	131.206
natural gas	10.330.564	11.982.991	12.957.446	15.219.275	17.034.548	20.457.793	21.607.635
lpg	9.521	759	12.673	12.908	33	0	0
naphtha	219.122	264.371	208.749	84.481	13.453	11.441	10.606

Net Calorific Values

	Net Calorific Value (NCVi,y) [GJ/t]						
	2002	2003	2004	2005	2006	2007	2008
hard coal	17,6	21,0	22,5	21,1	22,0	22,3	22,2
lignite	7,5	7,5	7,6	5,9	6,9	6,9	6,8

VCS Project Description Template

fuel oil	40,1	40,1	39,9	40,2	40,2	39,9	39,7
diesel oil	42,9	43,3	42,4	42,8	42,7	43,1	42,4
natural gas	40,8	39,1	38,0	38,6	37,0	36,7	36,6
lpg	46,2	44,1	45,9	46,1	0,0	0,0	0,0
naphtha	45,0	40,0	44,1	44,4	43,9	43,2	44,6

Operating Margin Emission Factor

	2002	2003	2004	2005	2006	2007	2008
EF_{grid,OM,y} (tCO₂/MWh)	0,721	0,657	0,672	0,638	0,657	0,663	0,664

EF_{Grid,OM} (tCO₂/MWh)	0,66149
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Details of BM Calculation

Recently Commissioned Power Plants

Company	Facility	Index	Installed Capacity (MW)	Commissioning Date	Energy Production (MWh)*	CO ₂ EF (ton/TJ)	tCO ₂ /MWh *	Facility Energy Efficiency	BM Emission Factor tCO ₂ /MWh	
MB Şeker Nişasta San. A.Ş. (Sultanhanı)	N.GAS	ANG	8,80	2008	60,0	54,3	0,20	60,0%	0,43	2
AKSA ENERJİ (Antalya)	N.GAS	ANG	183,80	2008	1.290,0	54,3	0,20	60,0%	0,43	50
AKSA ENERJİ (Manisa)	N.GAS	ANG	52,38	2008	370,0	54,3	0,20	60,0%	0,43	10
ANTALYA ENERJİ (ilave)	N.GAS	ANG	17,46	2008	122,3	54,3	0,20	60,0%	0,43	5
ATAÇ İNŞAAT SAN. A.S.B. (ANTALYTA)	N.GAS	ANG	5,40	2008	37,0	54,3	0,20	60,0%	0,43	1
BAHÇIVAN GIDA (LÜLEBURGAZ)	N.GAS	ANG	1,17	2008	8,0	54,3	0,20	60,0%	0,43	3
CAN ENERJİ (Çorlu - Tekirdağ) (İlave)	N.GAS	ANG	52,38	2008	304,2	54,3	0,20	60,0%	0,43	13
FOUR SEASONS OTEL (ATİK PASHA TUR A.Ş.)	N.GAS	ANG	1,17	2008	7,0	54,3	0,20	60,0%	0,43	3
FRITOLAY GIDA San. Ve TİC. A.Ş. (İlave)	N.GAS	ANG	0,60	2008	4,0	54,3	0,20	60,0%	0,43	1
ITC-KA Enerji Üretim A.Ş. (Mamak) (İlave)	RENEW.+ WASTES	AR	14,13	2008	107,0	-	-	na	-	0
KARKEY (SİLOPİ-5) (154kV) (İlave)	F.OIL	AF	14,78	2008	103,2	75,5	0,27	46,0%	1,26	13
MELİKE TEKSTİL (GAZİANTEP)	N.GAS	ANG	1,58	2008	11,0	54,3	0,20	60,0%	0,43	4
MİSİS APRE TEKSTİL BOYA EN. SAN.	N.GAS	ANG	2,00	2008	14,0	54,3	0,20	60,0%	0,43	6

VCS Project Description Template

MODERN ENERJİ (LİLEBURGAZ)	N.GAS	ANG	13,40	2008	94,1	54,3	0,20	60,0%	0,43	4
ORTADOĞU ENERJİ (ODA YERİ) (Eyip/İST.)	RENEW.+ WASTES	AR	2,83	2008	22,0	-	-	na	-	0
POLAT TURZ. (POLAT RENAISSANCE İST.OT.)	N.GAS	ANG	1,60	2008	11,0	54,3	0,20	60,0%	0,43	4
SARAYKÖY JEOTERMAL (Denizli)	GEOTHER MAL	AH	6,85	2008	50,0	-	-	na	-	0
YILDIZ SUNTA (Uzunçiftlik- Köseköy)(Düzelt)	N.GAS	ANG	22,63	2008	146,5	54,3	0,20	60,0%	0,43	6
SÖNMEZ Elektrik (İlave)	N.GAS	ANG	8,73	2008	67,3	54,3	0,20	60,0%	0,43	2
ALP ELEKTRİK TINAZTEPE / ANTALYA	HYDRO	AH	7,69	2008	29,0	-	-	na	-	0
CANSU ELEKTRİK (Murgul/Artvin)	HYDRO	AH	9,18	2008	47,0	-	-	na	-	0
ÇELDERE ELK. (ÇALDERE HES)	HYDRO	AH	8,74	2008	35,0	-	-	na	-	0
DAREN HES ELEKTRİK	HYDRO	AH	49,70	2008	182,0	-	-	na	-	0
DEĞİRMENÜSTÜ EN. (Kahramanmaraş)	HYDRO	AH	25,70	2008	69,0	-	-	na	-	0
GÖZEDE HES (TEMSA ELEKTRİK) Bursa	HYDRO	AH	2,40	2008	10,0	-	-	na	-	0
H.G.M. Enerji (KEKLİCEK HES)	HYDRO	AH	8,67	2008	18,0	-	-	na	-	0
HAMZALI HES (TURKON MNG ELEKTRİK)	HYDRO	AH	16,70	2008	117,0	-	-	na	-	0
HİDRO KNT. (YUKARI MANAHOZ Reg. Ve Hes)	HYDRO	AH	22,40	2008	79,0	-	-	na	-	0
İÇ-EN ELK. (ÇALKIŞLA REG. Ve HES)	HYDRO	AH	7,66	2008	18,0	-	-	na	-	0
KALEN ENERJİ (KALEN II Reg. Ve Hes)	HYDRO	AH	15,65	2008	50,0	-	-	na	-	0
MARAŞ ENERJİ (FIRNIS Reg. Ve HES)	HYDRO	AH	7,22	2008	36,0	-	-	na	-	0
SARMAŞIK I HES (FETAŞ FETHİYE ENERJİ)	HYDRO	AH	21,04	2008	96,0	-	-	na	-	0
SARMAŞIK II HES (FETAŞ FETHİYE ENERJİ)	HYDRO	AH	21,58	2008	108,0	-	-	na	-	0
TORUL	HYDRO	EH	105,60	2008	322,0	-	-	na	-	0

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YEŞİL ENERJİ ELEKTRİK (TAYFUN HES)	HYDRO	AH	0,82	2008	5,0	-	-	na	-	0
ZORLU ENERJİ (MERCAN)	HYDRO	AH	1,28	2008	4,9	-	-	na	-	0
BAKİ ELEKTRİK ŞAMLI RÜZGAR	WIND	AR	21,00	2008	104,0	-	-	na	-	0
DATÇA RES	WIND	AR	8,10	2008	24,0	-	-	na	-	0
ERTÜRK ELEKTRİK Çatalca RES	WIND	AR	60,00	2008	210,0	-	-	na	-	0
INNORES ELEKTRİK Yuntdağ RÜZG. (Aliğa)	WIND	AR	42,50	2008	161,0	-	-	na	-	0
LODOS RES (Taşoluk)	WIND	AR	24,00	2008	85,0	-	-	na	-	0
SAYALAR RÜZGAR	WIND	AR	30,60	2008	97,0	-	-	na	-	0
SEBENOBA (Deniz Elk.)	WIND	AR	31,20	2008	100,0	-	-	na	-	0
HABAŞ (Aliğa - İlave)	N.GAS	ANG	9,10	2007	72,8	54,3	0,20	60,0%	0,43	3
MODERN ENERJİ	N.GAS	ANG	5,20	2007	38,7	54,3	0,20	60,0%	0,43	1
ARENKO	N.GAS	ANG	0,10	2007	0,8	54,3	0,20	60,0%	0,43	0
ALTINMARKA GIDA	N.GAS	ANG	0,10	2007	0,8	54,3	0,20	60,0%	0,43	0
TEKBOY ENERJİ	N.GAS	ANG	0,10	2007	0,7	54,3	0,20	60,0%	0,43	0
VELSAN AKRILIK	N.GAS	ANG	0,10	2007	0,6	54,3	0,20	60,0%	0,43	0
Acıbaden Sağlık Hiz. Ve Tic. A.Ş. / Kadıköy	N.GAS	ANG	0,50	2007	4,0	54,3	0,20	60,0%	0,43	1
Acıbaden Sağlık Hiz. Ve Tic. A.Ş. / Kozyatağı	N.GAS	ANG	0,60	2007	5,0	54,3	0,20	60,0%	0,43	2
Acıbaden Sağlık Hiz. Ve Tic. A.Ş. / Bursa	N.GAS	ANG	1,30	2007	11,0	54,3	0,20	60,0%	0,43	4
AKATEKS	N.GAS	ANG	1,80	2007	14,0	54,3	0,20	60,0%	0,43	6
FLOKSER TEKSTİL / Poliser Tesisi	N.GAS	ANG	2,10	2007	17,0	54,3	0,20	60,0%	0,43	7
FLOKSER TEKSTİL / Sütser Tesisi	N.GAS	ANG	2,10	2007	17,0	54,3	0,20	60,0%	0,43	7
FRITOLAY GIDA	N.GAS	ANG	0,50	2007	4,0	54,3	0,20	60,0%	0,43	1
KIVANÇ TEKSTİL	N.GAS	ANG	3,90	2007	33,0	54,3	0,20	60,0%	0,43	1
KİL-KAN Kıl San. Ve Tic	N.GAS	ANG	3,20	2007	25,0	54,3	0,20	60,0%	0,43	1
SÜPERBOY BOYA SAN.	N.GAS	ANG	1,00	2007	8,0	54,3	0,20	60,0%	0,43	3
SWISS OTEL	N.GAS	ANG	1,60	2007	11,0	54,3	0,20	60,0%	0,43	4
TAV Esenboğa	N.GAS	ANG	3,90	2007	33,0	54,3	0,20	60,0%	0,43	1
NUH ENERJİ-2	N.GAS	ANG	73,00	2007	514,0	54,3	0,20	60,0%	0,43	23
AKTEKS	F.OIL	AF	0,80	2007	5,4	75,5	0,27	46,0%	1,26	6
UŞAK ŞEKER	LIGNITE	AL	1,70	2007	10,3	90,9	0,33	39,0%	1,20	1
BOĞAZLIYAN ŞEKER	N.GAS+N APHTA	ANG	16,40	2007	102,6	54,3	0,20	60,0%	0,43	4
KARTONSAN	N.GAS+N	ANG	5,00	2007	40,0	54,3	0,20	60,0%	0,43	1

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	APHTHA									
ESKİŞEHİR END. ENERJİ	N.GAS+N APHTHA	ANG	3,50	2007	26,8	54,3	0,20	60,0%	0,43	1
ESKİŞEHİR ŞEKER	N.GAS+N APHTHA	ANG	2,90	2007	18,1	54,3	0,20	60,0%	0,43	7
İGSAŞ	N.GAS+N APHTHA	ANG	2,20	2007	15,2	54,3	0,20	60,0%	0,43	6
DESA	N.GAS+N APHTHA	ANG	0,70	2007	5,6	54,3	0,20	60,0%	0,43	2
DENTAŞ	N.GAS+N APHTHA	ANG	0,30	2007	2,3	54,3	0,20	60,0%	0,43	1
SÜPER FİLMCİLİK	N.GAS+N APHTHA	ANG	0,10	2007	0,8	54,3	0,20	60,0%	0,43	0
ATAER ENERJİ	N.GAS+N APHTHA	ANG	0,10	2007	0,6	54,3	0,20	60,0%	0,43	0
BİL ENERJİ	N.GAS+N APHTHA	ANG	0,10	2007	0,7	54,3	0,20	60,0%	0,43	0
ITC-KA	RENEW.+ WASTES	AR	1,40	2007	11,1	-	-	na	-	0
BİS ENERJİ Bursa ilave	N.GAS	ANG	43,00	2007	354,8	54,3	0,20	60,0%	0,43	15
ALİAĞA ÇAKMAKTEPE	N.GAS	ANG	34,80	2007	278,0	54,3	0,20	60,0%	0,43	12
BİS ENERJİ Bursa DÜZELTİLME	N.GAS	ANG	28,30	2007	233,5	54,3	0,20	60,0%	0,43	10
BİS ENERJİ Bursa ilave	N.GAS	ANG	48,00	2007	396,1	54,3	0,20	60,0%	0,43	17
BOSEN ENERJİ	N.GAS	ANG	142,80	2007	1.071,0	54,3	0,20	60,0%	0,43	40
SAYENERJİ ELEKTRİK	N.GAS	ANG	5,90	2007	47,0	54,3	0,20	60,0%	0,43	2
T ENERJİ ÜRETİM A.Ş.	N.GAS	ANG	1,60	2007	13,0	54,3	0,20	60,0%	0,43	5
ZORLU ENERJİ Kayseri	N.GAS	ANG	7,20	2007	55,0	54,3	0,20	60,0%	0,43	2
SİİRT	F.OIL	AF	25,60	2007	190,0	75,5	0,27	46,0%	1,26	23
MARDİN KIZILTEPE	F.OIL	AF	34,10	2007	250,0	75,5	0,27	46,0%	1,26	3
KAREN	F.OIL	AF	24,30	2007	180,0	75,5	0,27	46,0%	1,26	22
İDİL 2 (PS3 A-2)	F.OIL	AF	24,40	2007	180,0	75,5	0,27	46,0%	1,26	22
BORÇKA HES	HYDRO	EH	300,60	2007	1.039,0	-	-	na	-	0
TEKTUĞ (KEBAN DERESİ)	HYDRO	AH	5,00	2007	32,0	-	-	na	-	0
YPM Ener. Yat. A.Ş. Altıntepe Hidro	HYDRO	AH	4,00	2007	18,0	-	-	na	-	0
YPM Ener. Yat. A.Ş. Beypınar Hidro	HYDRO	AH	3,60	2007	18,0	-	-	na	-	0
YPM Ener. Yat. A.Ş. Konak Hidro	HYDRO	AH	4,00	2007	19,0	-	-	na	-	0
KURTEKS Tekstil (KARASU HES - Andırın)	HYDRO	AH	2,40	2007	19,0	-	-	na	-	0
ISKUR TEKSTİL (SULEYMANLI HES)	HYDRO	AH	4,60	2007	18,0	-	-	na	-	0
ÖZGÜR ELK. AŞ.	HYDRO	AH	6,30	2007	27,0	-	-	na	-	0

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(K.MARAŞ) (Tahta)						-				
ÖZGÜR ELK. AŞ. (K.MARAŞ) (Tahta) İlave	HYDRO	AH	6,30	2007	27,0	-	-	na	-	0
ANEMON EN: ELEK. ÜRETİM A.Ş.	WIND	AR	8,00	2007	24,2	-	-	na	-	0
ANEMON EN: ELEK. ÜRETİM A.Ş. İlave	WIND	AR	15,20	2007	46,0	-	-	na	-	0
ANEMON EN: ELEK. ÜRETİM A.Ş. İlave	WIND	AR	7,20	2007	21,8	-	-	na	-	0
BURGAZ RES (Doğal Enerji Üretim A.Ş.)	WIND	AR	4,00	2007	12,9	-	-	na	-	0
BURGAZ RES (Doğal Enerji Üretim A.Ş.)	WIND	AR	10,90	2007	35,1	-	-	na	-	0
DENİZ ELEK. ÜRETİM Ltd. Şti. (Karakurt)	WIND	AR	10,80	2007	28,0	-	-	na	-	0
MARE MMANASTIR RÜZGAR ENERJİ	WIND	AR	11,2	2007	36,9	-	-	na	-	0
MARE MMANASTIR RÜZGAR ENERJİ	WIND	AR	20,00	2007	65,8	-	-	na	-	0
EKOTEN TEKSTİL GR-I	N.GAS	ANG	1,93	16.02.2006	14,0	54,3	0,20	60,0%	0,43	6
ERAK GİYİM GR-I	N.GAS	ANG	1,37	22.02.2006	10,0	54,3	0,20	60,0%	0,43	4
ALARKO ALTEK GR-III	N.GAS	ANG	21,89	23.02.2006	158,3	54,3	0,20	60,0%	0,43	6
AYDIN ÖRME GR-I	N.GAS	ANG	7,52	25.02.2006	60,0	54,3	0,20	60,0%	0,43	2
NUH ENERJİ-2 GR II	N.GAS	ANG	26,08	02.03.2006	180,1	54,3	0,20	60,0%	0,43	7
MARMARA ELEKTRİK (Çorlu) GR-I	N.GAS	ANG	8,73	13.04.2006	63,0	54,3	0,20	60,0%	0,43	2
MARMARA PAMUK (Çorlu) GR-I	N.GAS	ANG	8,73	13.04.2006	63,0	54,3	0,20	60,0%	0,43	2
ENTEK (Köseköy) GR IV	N.GAS	ANG	47,62	14.04.2006	391,3	54,3	0,20	60,0%	0,43	17
ELSE TEKSTİL (Çorlu) GR I-II	N.GAS	ANG	3,16	15.04.2006	25,0	54,3	0,20	60,0%	0,43	1
BARES IX GRUP	WIND	AR	13,50	20.04.2006	47,3	-	-	na	-	0
SÖNMEZ ELEKTRİK (Çorlu) GR I - II	N.GAS	ANG	17,46	03.05.2006	126,0	54,3	0,20	60,0%	0,43	5
DENİZLİ ÇİMENTO (DÜZELTME)	N.GAS	ANG	0,45	04.05.2006	3,2	54,3	0,20	60,0%	0,43	1
MENDERES ELEKTRİK GR I	GEOTHER MAL	AH	7,95	10.05.2006	56,0	-	-	na	-	0
KASTAMONU ENTEGRE BALIKSİR GR-I	N.GAS	ANG	7,52	24.05.2006	54,0	54,3	0,20	60,0%	0,43	2
BARES X. VE XX. GRUPLAR	WIND	AR	16,50	26.05.2006	57,8	-	-	na	-	0
BOZ ENERJİ GR-I	N.GAS	ANG	8,73	09.06.2006	70,0	54,3	0,20	60,0%	0,43	3

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ADANA ATIK SU ARITMA TESİSİ	RENEW.+ WASTES	AR	0,80	09.06.2006	6,0	-	-	na	-	0
AMYLUM NİŞASTA (ADANA)	N.GAS	ANG	14,25	09.06.2006	34,0	54,3	0,20	60,0%	0,43	1
ŞİK MAKAS (Çorlu) GR-I	N.GAS	ANG	1,58	22.06.2006	13,0	54,3	0,20	60,0%	0,43	5
ELBİSTAN B GR-III	LIGNITE	EL	360,00	23.06.2006	2.340,0	90,9	0,33	39,0%	0,93	2.1
ANTALYA ENERJİ GR I - II - III - IV	N.GAS	ANG	34,92	29.06.2006	245,0	54,3	0,20	60,0%	0,43	10
HAYAT TEM. VE SAĞLIK GR I - II	N.GAS	ANG	15,04	30.06.2006	108,0	54,3	0,20	60,0%	0,43	4
EKOLOJİK EN. (Kemerburgaz) GR I	RENEW.+ WASTES	AR	0,98	31.07.2006	6,0	-	-	na	-	0
EROĞLU GİYİM (Çorlu) GR-I	N.GAS	ANG	1,17	01.08.2006	9,0	54,3	0,20	60,0%	0,43	3
CAM İŞ ELEKTRİK (Mersin) GR I	N.GAS	ANG	126,10	13.09.2006	1.008,0	54,3	0,20	60,0%	0,43	43
ELBİSTAN B GR II	LIGNITE	EL	360,00	17.09.2006	2.340,0	90,9	0,33	39,0%	0,93	2.1
YILDIZ ENT. AĞAÇ (Kocaeli) GR I	N.GAS	ANG	6,18	21.09.2006	40,0	54,3	0,20	60,0%	0,43	1
ÇERKEZKÖY ENERJİ GR I	N.GAS	ANG	49,16	06.10.2006	390,0	54,3	0,20	60,0%	0,43	10
ENTEK (Köseköy) GR V	N.GAS	ANG	37,00	03.11.2006	304,0	54,3	0,20	60,0%	0,43	13
ITC-KA EN. MAMAK TOP.M. GR I-II-III	RENEW.+ WASTES	AR	4,24	03.11.2006	30,0	-	-	na	-	0
ELBİSTAN B GRUP IV	LIGNITE	EL	360,00	13.11.2006	2.340,0	90,9	0,33	39,0%	0,93	2.1
MARE MANASTIR RÜZGAR (X GRUP)	WIND	AR	8,00	08.12.2006	25,0	-	-	na	-	0
ÇİRAĞAN SARAYI GR I	N.GAS	ANG	1,32	01.12.2006	11,0	54,3	0,20	60,0%	0,43	4
ERTÜRK ELEKTRİK Tepe RES GR I	WIND	AR	0,85	22.12.2006	2,0	-	-	na	-	0
AKMAYA (Lüleburgaz) GR I	N.GAS	ANG	6,91	23.12.2006	50,0	54,3	0,20	60,0%	0,43	2
BURGAZ (Lüleburgaz) GR I	N.GAS	ANG	6,91	23.12.2006	54,0	54,3	0,20	60,0%	0,43	2
SEYHAN I-II	HYDRO	EH	0,30	20.02.2006	1,7	-	-	na	-	0
ŞANLIURFA GR I-II	HYDRO	EH	51,80	01.03.2006	124,0	-	-	na	-	0
BEREKET ENERJİ GÖKYAR HES 3 Grup	HYDRO	AH	11,62	05.05.2006	43,3	-	-	na	-	0
MOLU EN. Zamantı Bahçelik GR I - II	HYDRO	AH	4,22	31.05.2006	16,7	-	-	na	-	0
SU ENERJİ BALIKESİR GR I - II	HYDRO	AH	4,60	27.06.2006	20,7	-	-	na	-	0
BEREKET EN. MENTAŞ REG. GR I - II	HYDRO	AH	26,60	31.07.2006	108,7	-	-	na	-	0
EKİN (Başaran Hes) (Nazilli)	HYDRO	AH	0,60	11.08.2006	4,5	-	-	na	-	0
ERE (Sugözü Reg. Ve Kızıldüz HES)	HYDRO	AH	15,43	08.09.2006	31,6	-	-	na	-	0
ERE (AKSU REG.	HYDRO	AH	14,00	16.11.2006	26,7	-	-	na	-	0

VCS Project Description Template

VE ŞAHMALLAR HES)										
TEKTUĞ (KALEALTI) GR I - II	HYDRO	AH	15,00	30.11.2006	52,0	-	-	na	-	0
BEREKET EN. MENTAŞ REG. GR III	HYDRO	AH	13,30	13.12.2006	54,4	-	-	na	-	0
BOSEN GR-III	N.GAS	ANG	51,02	30.12.2005	372,8	54,3	0,20	60,0%	0,43	10
KARKEY (SİLOPİ-4) GR-V	F.OIL	AF	6,75	23.12.2005	51,9	75,5	0,27	46,0%	1,26	6
AKÇA ENERJİ GR-III	N.GAS+N APHTA	ANG	8,73	14.12.2005	65,5	54,3	0,20	60,0%	0,43	2
KAHRAMANMARAŞ KAĞIT GR-I	IMPORTE D COAL	AHC	6,00	08.12.2005	45,0	92,8	0,33	39,0%	0,76	3
PAK GIDA	N.GAS	ANG	5,67	07.12.2005	45,0	54,3	0,20	60,0%	0,43	1
KORUMA KLOR GR I-II-III	N.GAS	ANG	9,60	03.12.2005	77,0	54,3	0,20	60,0%	0,43	3
İÇDAŞ ÇELİK GR-I	IMPORTE D COAL	AHC	135,00	30.11.2005	1.080,0	92,8	0,33	39,0%	0,76	8
KÜÇÜKÇALIK TEKSTİL GR I-II-III-IV	N.GAS	ANG	8,00	27.11.2005	64,0	54,3	0,20	60,0%	0,43	2
ZORLU ENERJİ YALOVA GR I-II	N.GAS	ANG	15,93	26.11.2005	122,0	54,3	0,20	60,0%	0,43	5
HABAŞ ALİAĞA GR-V	N.GAS	ANG	23,00	24.11.2005	184,0	54,3	0,20	60,0%	0,43	8
GRANİSER GRANİT GR-I	N.GAS	ANG	5,50	14.11.2005	42,0	54,3	0,20	60,0%	0,43	1
MOSB GR I-II-III-IV-V-VI-VII	N.GAS	ANG	84,83	11.11.2005	434,0	54,3	0,20	60,0%	0,43	18
AK ENERJİ(K.paşa) GR- III	N.GAS	ANG	40,00	09.11.2005	256,9	54,3	0,20	60,0%	0,43	1
ZORLU ENERJİ KAYSERİ GR-IV	N.GAS	ANG	38,63	26.10.2005	294,9	54,3	0,20	60,0%	0,43	12
ALTEK ALARKO GR I-II	N.GAS	ANG	60,10	14.10.2005	420,0	54,3	0,20	60,0%	0,43	18
AYKA TEKSTİL GR-I	N.GAS	ANG	5,50	24.09.2005	40,0	54,3	0,20	60,0%	0,43	1
HABAŞ ALİAĞA GR IV	N.GAS	ANG	44,62	21.09.2005	357,0	54,3	0,20	60,0%	0,43	15
EVYAP GR I-II	N.GAS	ANG	5,12	27.08.2005	30,0	54,3	0,20	60,0%	0,43	1
ÇEBİ ENERJİ BT	N.GAS	ANG	21,00	27.08.2005	164,7	54,3	0,20	60,0%	0,43	7
CAN ENERJİ GR-I	N.GAS	ANG	3,90	25.08.2005	28,0	54,3	0,20	60,0%	0,43	1
NOREN ENERJİ GR-I	N.GAS	ANG	8,73	24.08.2005	70,0	54,3	0,20	60,0%	0,43	3
ÇEBİ ENERJİ GT	N.GAS	ANG	43,37	23.08.2005	340,1	54,3	0,20	60,0%	0,43	14
YAMULA GRUP I-II	HYDRO	EH	100,00	31.07.2005	422,0	-	-	na	-	0
ZORLU ENERJİ KAYSERİ GR-I-II-III	N.GAS	ANG	149,87	22.07.2005	1.144,1	54,3	0,20	60,0%	0,43	49
BEREKET EN. (DALAMAN) GR XIII-XIV-XV	HYDRO	AH	7,50	15.07.2005	35,8	-	-	na	-	0
ETİ MAD.(BAN.ASİT)GR	RENEW.+ WASTES	AR	11,50	15.07.2005	88,0	-	-	na	-	0

VCS Project Description Template

-I										
ZEYNEP GİYİM SAN. GR-I	N.GAS	ANG	1,17	07.07.2005	9,0	54,3	0,20	60,0%	0,43	3
KARKEY (SİLOPİ-4) GR-IV	F.OIL	AF	6,15	30.06.2005	47,2	75,5	0,27	46,0%	1,26	5
AKBAŞLAR GR- II(İZOLE)	N.GAS	ANG	9,00	24.06.2005	71,3	54,3	0,20	60,0%	0,43	3
MODERN ENERJİ (DG) GR-III	N.GAS	ANG	8,38	14.06.2005	61,1	54,3	0,20	60,0%	0,43	2
MODERN ENERJİ (DG+LPG) GR-II	N.GAS+LP G	ANG	7,68	13.06.2005	56,0	54,3	0,20	60,0%	0,43	2
MODERN ENERJİ (DG+LPG) GR-II (DÜZELTME)	N.GAS+LP G	ANG	4,50	13.06.2005	32,8	54,3	0,20	60,0%	0,43	1
MURATLI GR I-II	HYDRO	EH	115,00	03.06.2005	444,0	-	-	na	-	0
HABAŞ ALİAĞA GR III	N.GAS	ANG	44,62	02.06.2005	356,9	54,3	0,20	60,0%	0,43	13
HAYAT KAĞIT GR-I	N.GAS	ANG	7,53	27.05.2005	56,0	54,3	0,20	60,0%	0,43	2
TEZCAN GALVANİZ GR I-II	N.GAS	ANG	3,66	27.05.2005	29,0	54,3	0,20	60,0%	0,43	1
YONGAPAN(KAST. ENTG) GR-II	N.GAS	ANG	5,20	25.05.2005	35,8	54,3	0,20	60,0%	0,43	1
NUH ENERJİ-2 GR I	N.GAS	ANG	46,95	24.05.2005	319,7	54,3	0,20	60,0%	0,43	13
İÇTAŞ ENERJİ (Yukarı Mercan) GR I-II	HYDRO	AH	14,19	22.05.2005	44,0	-	-	na	-	0
AK ENERJİ(K.paşa) GR I-II	N.GAS	ANG	87,20	30.04.2005	560,1	54,3	0,20	60,0%	0,43	24
TEKTUĞ (Kargılık) GR I-II	HYDRO	AH	23,90	25.04.2005	83,0	-	-	na	-	0
SUNJÜT(RES) GR I- II	WIND	AR	1,20	23.04.2005	2,0	-	-	na	-	0
KAREGE GR IV-V	N.GAS	ANG	18,06	07.04.2005	141,9	54,3	0,20	60,0%	0,43	6
BİS ENERJİ GR VII	N.GAS	ANG	43,70	18.03.2005	287,6	54,3	0,20	60,0%	0,43	12
ÇAN GR I (EÜAŞ)	LIGNITE	EL	160,00	15.03.2005	1.040,0	90,9	0,33	39,0%	0,93	90
ÇAN GR I (EÜAŞ)	LIGNITE	EL	160,00	15.02.2005	1.040,0	90,9	0,33	39,0%	0,93	90
ELBİSTAN-B GR I (EÜAŞ)	LIGNITE	EL	360,00	15.02.2005	2.340,0	90,9	0,33	39,0%	0,93	2.1
ENTEK ELK.A.Ş.KOÇ ÜNİ.GR I-II	N.GAS	ANG	2,33	07.02.2005	19,0	54,3	0,20	60,0%	0,43	8
BAYDEMİRLER GR IV-V-VI	N.GAS	ANG	6,21	04.02.2005	49,3	54,3	0,20	60,0%	0,43	2
MERCEDES BENZ TURK GR I-II-III-IV	N.GAS	ANG	8,28	04.02.2005	68,0	54,3	0,20	60,0%	0,43	2
METEM ENERJİ (Hacııramat) GR I-II	N.GAS	ANG	7,83	29.01.2005	58,0	54,3	0,20	60,0%	0,43	2
METEM ENERJİ (Peliklik) GR I-II-III	N.GAS	ANG	11,75	29.01.2005	89,0	54,3	0,20	60,0%	0,43	3
ALTINMARKA GIDA GR I-II-III	N.GAS	ANG	3,60	17.12.2004	28,8	54,3	0,20	60,0%	0,43	1
KARKEY-II 3+3 DGM	FUEL-OIL	AF	54,30	12.11.2004	370,0	75,5	0,27	46,0%	1,26	40
STANDART PROFİL 3 GM	N.GAS	ANG	6,74	22.10.2004	49,2	54,3	0,20	60,0%	0,43	2

VCS Project Description Template

HABAŞ ALİAĞA GRUP I-II	N.GAS	ANG	89,23	08.10.2004	713,7	54,3	0,20	60,0%	0,43	3
AYEN OSTİM ENERJİ ÜRETİM(BT)	N.GAS	ANG	9,89	01.10.2004	84,0	54,3	0,20	60,0%	0,43	3
KOMBASSAN KAĞ. MATBAA GIDA	N.GAS	ANG	5,50	24.09.2004	35,7	54,3	0,20	60,0%	0,43	1
BEREKET EN.(Feslek Hes) Gr-1-2	HYDRO	AH	9,48	05.08.2004	41,0	-	-	na	-	0
ÇELİK ENERJİ ÜR.ŞTİ. 2 GM	N.GAS	ANG	2,42	09.07.2004	19,0	54,3	0,20	60,0%	0,43	8
BESLER GR-2, BT (5,2+7,5)	N.GAS	ANG	12,70	07.07.2004	95,3	54,3	0,20	60,0%	0,43	4
ŞAHİNLER ENERJİ 1 GM	N.GAS	ANG	3,20	29.06.2004	24,9	54,3	0,20	60,0%	0,43	1
ENERJİ-SA ADANA 1 BT	NAPHTA	AN	49,77	23.06.2004	373,3	69,3	0,25	60,0%	0,77	28
BİS ENERJİ 2 GT	N.GAS	ANG	73,04	16.06.2004	602,6	54,3	0,20	60,0%	0,43	20
AYEN OSTİM ENERJİ ÜRETİM	N.GAS	ANG	31,08	11.06.2004	264,1	54,3	0,20	60,0%	0,43	1
KOMBASSAN KAĞIT GIDA VE TEKS	N.GAS	ANG	5,50	09.06.2004	38,1	54,3	0,20	60,0%	0,43	1
GÜL ENERJİ GR-II	FUEL-OIL	AF	12,50	03.06.2004	93,8	75,5	0,27	46,0%	1,26	1
TEKBOY TEKSTİL 1 GM	N.GAS	ANG	2,25	18.05.2004	16,0	54,3	0,20	60,0%	0,43	7
ÇOLAKOĞLU(KAP ASİTE ARTIRIMI)	IMPORTE D COAL	AHC	45,00	05.05.2004	347,8	92,8	0,33	39,0%	0,76	20
İSKUR TEKSTİL(SÜLEYMANLI) GR I-II	HYDRO	AR	4,60	28.04.2004	17,9	-	-	na	-	0
ELTA ELK(DODURGA) GR-I-II-III-IV	HYDRO	AR	4,14	26.04.2004	12,3	-	-	na	-	0
TANRIVERDİ 4 GM	N.GAS	ANG	4,66	24.03.2004	38,7	54,3	0,20	60,0%	0,43	1
ERE(BİR KAPILI HES) GRUP-I	HYDRO	AR	48,50	11.03.2004	170,5	-	-	na	-	0
ATATEKS 2 GM	N.GAS	ANG	5,63	20.02.2004	45,0	54,3	0,20	60,0%	0,43	1
ENTEK GR-IV	N.GAS+N APHTA	ANG	31,13	12.02.2004	233,5	54,3	0,20	60,0%	0,43	10
ANKARA D.G.(BAYMİNA) GR-I-II-III	N.GAS	ANG	798,00	08.01.2004	6.500,0	54,3	0,20	60,0%	0,43	2.8

TOTAL

45.670,2

**Facility
Energy
Efficiency**

24.

Built Margin Emission Factor

	Total Electricity Generation (GWh)	Net Electricity Generation (GWh)	Total CO2 Emissions (Ton)	BM Values for each year (tCO2/MWh)
2008	4.835,6	4.624,6	889,9	0,19243
2007	5.785,8	5.537,6	1.614,2	0,29150

2006	11.218,1	10.788,2	7.019,7	0,65069
2005	13.615,5	13.070,1	6.958,0	0,53236
2004	10.215,2	9.833,4	3.587,7	0,36485

BM Emission Factor :	0,45764
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Summary of Grid Emission Factor Calculation

Parameter	SI Unit	Result
EFGGrid,OM,y	tCO2/MWh	0,661
EFGGrid,BM,y	tCO2/MWh	0,457
EFGGrid,CM,y	tCO2/MWh	0,559

ANNEX 4

EIA EXEMPTION LETTERS

1st Letter dated 03^d February 2006

The document below explains that: According to the 17th article of Regulation on Environmental Impact Assessment (Date: 16.12.2003, Number: 25318). Also it is not required for subject project to prepare an EIA Report in line with 17th article of Regulation on Environmental Assessment.

T.C. Çevre ve Orman Bakanlığı

T.C. ÇEVRE ve ORMAN BAKANLIĞI
ÇEVRESEL ETKİ DEĞERLENDİRMESİ VE PLANLAMA
GENEL MÜDÜRLÜĞÜ

#01247

Karar Tarihi : 3.../08/2006
Karar No :

ÇED GEREKLİ DEĞİLDİR BELGESİ

16.12.2003 tarih ve 25318 sayılı Resmi Gazete'de yayımlanarak yürürlüğe giren Çevresel Etki Değerlendirmesi Yönetmeliği'nin 17. maddesi gereğince; "Uzundere - Regülatörü ve HES (43,68 kW)" projesi hakkında "Çevresel Etki Değerlendirmesi Gerekli Değildir" kararı verilmiştir.

12 Ocak 2010

Osman TÜZÜN
Bakan
Genel Müdür

Proje Sahibi : Karadeniz Enerji Yatırım San. Tic. A.Ş.
Projenin Yeri : Rize İl, Çayeli İlçesi.

2nd Letter dated 2th January 2009

The document below explains that: As a run-off-river hydro project with an installed capacity of more than 0,5 MW, the project is investigated/assessed and the anticipated precautions in the Project Description File are found sufficient in accordance with the 17th article of Regulation on Environmental Impact Assessment (Date: 17.07.2008, Number: 26939). Also it is not required for subject project to prepare an EIA Report in line with 17th article of Regulation on Environmental Assessment.

**T.C.
RİZE VALİLİĞİ
İL ÇEVRE VE ORMAN MÜDÜRLÜĞÜ**

**Karar Tarihi:27.01.2009
Karar No :1**

ÇEVRESEL ETKİ DEĞERLENDİRME BELGESİ

17 Temmuz 2008 tarih ve 26939 sayılı Resmi Gazete'de yayımlanarak yürürlüğe giren Çevresel Etki Değerlendirmesi Yönetmeliğinin Ek-II listesinde yer alan "Kurulu gücü 0,5MW ve üzeri olan nehir tipi santraller" projesi ile ilgili olarak inceleme-değerlendirme yapılmış ve Proje Tanıtım Dosyasında çevresel etkilere karşı alınması öngörülen önlemler yeterli görülmüştür. Ayrıca ÇED Raporu hazırlanmasına gerek bulunmadığı tespit edilmiş olup, söz konusu projeye ÇED Yönetmeliğinin 17. maddesi gereğince Valiliğimizce "Çevresel Etki Değerlendirmesi Gerekli Değildir Kararı" verilmiştir.

ÇEVRESEL ETKİ DEĞERLENDİRME BELGESİ
ÇEVRESEL ETKİ DEĞERLENDİRME BELGESİ
Araştırma ve Değerlendirme
T.C. 20.016.497196
İlçe Kurulunun kararına göre edilen
ASLINA KAYITLI BELGE
Araştırma ve Değerlendirme İlçe Kurulunun kararına göre edilen
güncelleme gerektirmeyen kararname

Sabit KANDEMİR
İl Çevre ve Orman Müdürü

Proje Sahibi : Karadeniz Elektrik Üretim San.A.Ş.
Projenin Yeri : Rize İli, Çayeli İlçesi, Çataldere Köyü Mevkii.
Koordinat: 40°45'4"- 40°49'42"DOĞU 40°50'16"-40°54'04" KUZAY

01246
ÇEVRESEL ETKİ DEĞERLENDİRME
Araştırma ve Değerlendirme
T.C. 20.016.497196
Yazın ALINAC

12 Ocak 2009

ANNEX 5

POSITIVE EIA LETTER

Letter dated 01ST April 2010

The document below explains that: According to the 14th article of Regulation on Environmental Impact Assessment (Date: 17.07.2008, Number: 26939,) a Positive EIA decision was made.

	T.C. ÇEVRE ve ORMAN BAKANLIĞI ÇEVRESEL ETKİ DEĞERLENDİRMESİ VE PLANLAMA GENEL MÜDÜRLÜĞÜ	Karar Tarihi: 01.04/2010 Karar No : 1881
ÇED OLUMLU BELGESİ		
17.07.2008 tarih ve 26939 sayılı Resmi Gazete’de yayımlanarak yürürlüğe giren Çevresel Etki Değerlendirmesi Yönetmeliği’nin 14. maddesi gereğince; “Uzundere 1 Regülatörü ve HES (63,068 MWm)” projesi hakkında “Çevresel Etki Değerlendirmesi Olumlu” kararı verilmiştir.		
		 Fevzi İŞBİLİR Bakan a. Genel Müdür
Proje Sahibi Projenin Yeri	: Karadeniz Elektrik Üretim Sanayi A.Ş. : Rize İli, Çayeli İlçesi,	

